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20                         "operator_weights": {
21                             "<": 32.7,
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72                         }
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204             }
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622         }
623     ]
624 }

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622     }
623     ]
624     }
625     ]
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646                         "=": 0.025257134810090065
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679                     "pair_idx": 374,
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682                         "<": 0.0290681105107069,
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685                     }
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688                     "pair_idx": 438,
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690                     "weights": {

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693         "=": 0.025483369827270508
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695 },
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702         "=": 0.0291337538510561
703     }
704 }
705 ]
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713         "=": 33.0
714     },
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740                 "=": 0.02408851869404316
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751         "=": 33.7
752     },
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756             "dominant_op": ">",
757             "weights": {
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761     }
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766     "weights": {
767         "<": 0.025471076369285583,
768         ">": 0.028382526710629463,
769         "=": 0.018153272569179535
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776         "<": 0.025162290781736374,
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778         "=": 0.020395705476403236
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783 {
784     "index": 4,
785     "steepness": 19.993064880371094,
786     "operator_weights": {
787         "<": 33.4,
788         ">": 33.1,
789         "=": 33.5
790     },
791     "top_comparisons": [
792     {
793         "pair_idx": 312,
794         "dominant_op": "=",
795         "weights": {
796             "<": 0.026592504233121872,
797             ">": 0.02592904306948185,
798             "=": 0.029482288286089897
799         }
800     },
801     {
802         "pair_idx": 430,
803         "dominant_op": "<",
804         "weights": {
805             "<": 0.026585185900330544,
806             ">": 0.024637123569846153,
807             "=": 0.025915469974279404
808         }
809     },
810     {
811         "pair_idx": 475,
812         "dominant_op": ">",
813         "weights": {
814             "<": 0.0250706784427166,
815             ">": 0.030077198520302773,
816             "=": 0.021584702655673027
817         }
818     }
819 ]
820 },
821 {
822     "index": 5,
823     "steepness": 20.01475715637207,
824     "operator_weights": {
825         "<": 33.6,
826         ">": 34.2,
827         "=": 32.1
828     },

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829         "top_comparisons": [
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831                 "pair_idx": 142,
832                 "dominant_op": "=",
833                 "weights": {
834                     "<": 0.02611907199025154,
835                     ">": 0.028710462152957916,
836                     "=": 0.029112765565514565
837                 }
838             },
839             {
840                 "pair_idx": 124,
841                 "dominant_op": ">",
842                 "weights": {
843                     "<": 0.020887555554509163,
844                     ">": 0.02862139604985714,
845                     "=": 0.025669140741229057
846                 }
847             },
848             {
849                 "pair_idx": 153,
850                 "dominant_op": "<",
851                 "weights": {
852                     "<": 0.02712913602590561,
853                     ">": 0.02463679201900959,
854                     "=": 0.02247515320777893
855                 }
856             }
857         ]
858     },
859     {
860         "index": 6,
861         "steepness": 20.00912857055664,
862         "operator_weights": {
863             "<": 34.3,
864             ">": 32.7,
865             "=": 33.0
866         },
867         "top_comparisons": [
868             {
869                 "pair_idx": 479,
870                 "dominant_op": ">",
871                 "weights": {
872                     "<": 0.024133699014782906,
873                     ">": 0.03130073845386505,
874                     "=": 0.025021862238645554
875                 }
876             },
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878                 "pair_idx": 408,
879                 "dominant_op": "<",
880                 "weights": {
881                     "<": 0.03180775046348572,
882                     ">": 0.024859262630343437,
883                     "=": 0.022081099450588226
884                 }
885             },
886             {
887                 "pair_idx": 53,
888                 "dominant_op": "=",
889                 "weights": {
890                     "<": 0.024400800466537476,
891                     ">": 0.02387210913002491,
892                     "=": 0.025657009333372116
893                 }
894             }
895         ]
896     },
897     {

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898         "index": 7,
899         "steepness": 19.99560546875,
900         "operator_weights": {
901             "<": 33.4,
902             ">": 34.3,
903             "=": 32.3
904         },
905         "top_comparisons": [
906             {
907                 "pair_idx": 275,
908                 "dominant_op": "<",
909                 "weights": {
910                     "<": 0.028200846165418625,
911                     ">": 0.025956014171242714,
912                     "=": 0.027550065889954567
913                 }
914             },
915             {
916                 "pair_idx": 375,
917                 "dominant_op": ">",
918                 "weights": {
919                     "<": 0.02061467058956623,
920                     ">": 0.03637133911252022,
921                     "=": 0.01957716979086399
922                 }
923             },
924             {
925                 "pair_idx": 298,
926                 "dominant_op": "=",
927                 "weights": {
928                     "<": 0.027642525732517242,
929                     ">": 0.018996918573975563,
930                     "=": 0.0277139600366354
931                 }
932             }
933         ]
934     },
935     {
936         "index": 8,
937         "steepness": 20.000511169433594,
938         "operator_weights": {
939             "<": 33.7,
940             ">": 33.9,
941             "=": 32.4
942         },
943         "top_comparisons": [
944             {
945                 "pair_idx": 434,
946                 "dominant_op": "=",
947                 "weights": {
948                     "<": 0.024069616571068764,
949                     ">": 0.022618334740400314,
950                     "=": 0.024642426520586014
951                 }
952             },
953             {
954                 "pair_idx": 397,
955                 "dominant_op": "<",
956                 "weights": {
957                     "<": 0.024357976391911507,
958                     ">": 0.021719522774219513,
959                     "=": 0.024283356964588165
960                 }
961             },
962             {
963                 "pair_idx": 392,
964                 "dominant_op": "=",
965                 "weights": {
966                     "<": 0.027151362970471382,

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967         ">": 0.015233066864311695,
968         "=": 0.02796783670783043
969     }
970 }
971 ]
972 },
973 {
974     "index": 9,
975     "steepness": 19.998615264892578,
976     "operator_weights": {
977         "<": 33.3,
978         ">": 34.1,
979         "=": 32.6
980     },
981     "top_comparisons": [
982         {
983             "pair_idx": 148,
984             "dominant_op": "=",
985             "weights": {
986                 "<": 0.023016048595309258,
987                 ">": 0.02628115378320217,
988                 "=": 0.03173718973994255
989             }
990         },
991         {
992             "pair_idx": 187,
993             "dominant_op": ">",
994             "weights": {
995                 "<": 0.02608339861035347,
996                 ">": 0.029319679364562035,
997                 "=": 0.017235351726412773
998             }
999         },
1000         {
1001             "pair_idx": 294,
1002             "dominant_op": "<",
1003             "weights": {
1004                 "<": 0.029068436473608017,
1005                 ">": 0.024144399911165237,
1006                 "=": 0.01880233734846115
1007             }
1008         }
1009     ]
1010 },
1011 {
1012     "index": 10,
1013     "steepness": 20.008699417114258,
1014     "operator_weights": {
1015         "<": 32.0,
1016         ">": 33.8,
1017         "=": 34.2
1018     },
1019     "top_comparisons": [
1020         {
1021             "pair_idx": 229,
1022             "dominant_op": ">",
1023             "weights": {
1024                 "<": 0.02352042682468891,
1025                 ">": 0.025101428851485252,
1026                 "=": 0.01964917592704296
1027             }
1028         },
1029         {
1030             "pair_idx": 315,
1031             "dominant_op": "<",
1032             "weights": {
1033                 "<": 0.025317629799246788,
1034                 ">": 0.019736621528863907,
1035                 "=": 0.022828062996268272

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1036     }
1037     },
1038     {
1039         "pair_idx": 154,
1040         "dominant_op": "<",
1041         "weights": {
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1043             ">": 0.021777940914034843,
1044             "=": 0.02177913300693035
1045         }
1046     }
1047 ]
1048 },
1049 {
1050     "index": 11,
1051     "steepness": 19.99726676940918,
1052     "operator_weights": {
1053         "<": 33.0,
1054         ">": 33.8,
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1056     },
1057     "top_comparisons": [
1058         {
1059             "pair_idx": 99,
1060             "dominant_op": "=",
1061             "weights": {
1062                 "<": 0.025004899129271507,
1063                 ">": 0.020606625825166702,
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1065             }
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1069             "dominant_op": "<",
1070             "weights": {
1071                 "<": 0.029611486941576004,
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1073                 "=": 0.025054417550563812
1074             }
1075         },
1076         {
1077             "pair_idx": 71,
1078             "dominant_op": "<",
1079             "weights": {
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1081                 ">": 0.019526487216353416,
1082                 "=": 0.02472398802638054
1083             }
1084         }
1085     ]
1086 },
1087 {
1088     "index": 12,
1089     "steepness": 19.996681213378906,
1090     "operator_weights": {
1091         "<": 32.0,
1092         ">": 34.0,
1093         "=": 34.0
1094     },
1095     "top_comparisons": [
1096         {
1097             "pair_idx": 38,
1098             "dominant_op": "=",
1099             "weights": {
1100                 "<": 0.027392171323299408,
1101                 ">": 0.023377664387226105,
1102                 "=": 0.02905147336423397
1103             }
1104         },

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1105         {
1106             "pair_idx": 297,
1107             "dominant_op": "<",
1108             "weights": {
1109                 "<": 0.02706114575266838,
1110                 ">": 0.0230498518794775,
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1112             }
1113         },
1114         {
1115             "pair_idx": 63,
1116             "dominant_op": "<",
1117             "weights": {
1118                 "<": 0.027219140902161598,
1119                 ">": 0.023024937137961388,
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1122         }
1123     ]
1124 },
1125 {
1126     "index": 13,
1127     "steepness": 20.001445770263672,
1128     "operator_weights": {
1129         "<": 32.9,
1130         ">": 33.8,
1131         "=": 33.4
1132     },
1133     "top_comparisons": [
1134         {
1135             "pair_idx": 314,
1136             "dominant_op": ">",
1137             "weights": {
1138                 "<": 0.020558012649416924,
1139                 ">": 0.02845282293856144,
1140                 "=": 0.026259994134306908
1141             }
1142         },
1143         {
1144             "pair_idx": 261,
1145             "dominant_op": ">",
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1147                 "<": 0.025654073804616928,
1148                 ">": 0.025822700932621956,
1149                 "=": 0.02378268912434578
1150             }
1151         },
1152         {
1153             "pair_idx": 119,
1154             "dominant_op": "<",
1155             "weights": {
1156                 "<": 0.028412986546754837,
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1158                 "=": 0.02102350816130638
1159             }
1160         }
1161     ]
1162 },
1163 {
1164     "index": 14,
1165     "steepness": 20.00785255432129,
1166     "operator_weights": {
1167         "<": 34.8,
1168         ">": 32.4,
1169         "=": 32.8
1170     },
1171     "top_comparisons": [
1172         {
1173             "pair_idx": 184,

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1174         "dominant_op": "<",
1175         "weights": {
1176             "<": 0.026968514546751976,
1177             ">": 0.025964954867959023,
1178             "=": 0.024186523631215096
1179         }
1180     },
1181     {
1182         "pair_idx": 349,
1183         "dominant_op": "<",
1184         "weights": {
1185             "<": 0.03515053167939186,
1186             ">": 0.02687128446996212,
1187             "=": 0.014219586737453938
1188         }
1189     },
1190     {
1191         "pair_idx": 424,
1192         "dominant_op": ">",
1193         "weights": {
1194             "<": 0.026437491178512573,
1195             ">": 0.02664252743124962,
1196             "=": 0.020500004291534424
1197         }
1198     }
1199 ]
1200 },
1201 {
1202     "index": 15,
1203     "steepness": 19.995147705078125,
1204     "operator_weights": {
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1206         ">": 33.3,
1207         "=": 34.2
1208     },
1209     "top_comparisons": [
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1211             "pair_idx": 329,
1212             "dominant_op": "<",
1213             "weights": {
1214                 "<": 0.029656745493412018,
1215                 ">": 0.02169675938785076,
1216                 "=": 0.02962665818631649
1217             }
1218         },
1219         {
1220             "pair_idx": 8,
1221             "dominant_op": "<",
1222             "weights": {
1223                 "<": 0.02840079367160797,
1224                 ">": 0.019768888130784035,
1225                 "=": 0.027726663276553154
1226             }
1227         },
1228         {
1229             "pair_idx": 466,
1230             "dominant_op": "<",
1231             "weights": {
1232                 "<": 0.0304828193038702,
1233                 ">": 0.025056354701519012,
1234                 "=": 0.01944717951118946
1235             }
1236         }
1237     ]
1238 }
1239 ]
1240 },
1241 {
1242     "index": 2,

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1243     "n_sets": 16,
1244     "sets": [
1245         {
1246             "index": 0,
1247             "steepness": 20.00335121154785,
1248             "operator_weights": {
1249                 "<": 32.1,
1250                 ">": 33.5,
1251                 "=": 34.4
1252             },
1253             "top_comparisons": [
1254                 {
1255                     "pair_idx": 121,
1256                     "dominant_op": ">",
1257                     "weights": {
1258                         "<": 0.02784120850265026,
1259                         ">": 0.027864733710885048,
1260                         "=": 0.02177424728870392
1261                     }
1262                 },
1263                 {
1264                     "pair_idx": 465,
1265                     "dominant_op": ">",
1266                     "weights": {
1267                         "<": 0.025656046345829964,
1268                         ">": 0.02864198386669159,
1269                         "=": 0.021348614245653152
1270                     }
1271                 },
1272                 {
1273                     "pair_idx": 258,
1274                     "dominant_op": "=",
1275                     "weights": {
1276                         "<": 0.02324488013982773,
1277                         ">": 0.021921545267105103,
1278                         "=": 0.028914282098412514
1279                     }
1280                 }
1281             ]
1282         },
1283         {
1284             "index": 1,
1285             "steepness": 20.002246856689453,
1286             "operator_weights": {
1287                 "<": 33.1,
1288                 ">": 32.7,
1289                 "=": 34.2
1290             },
1291             "top_comparisons": [
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1293                     "pair_idx": 243,
1294                     "dominant_op": ">",
1295                     "weights": {
1296                         "<": 0.021723289042711258,
1297                         ">": 0.02344704046845436,
1298                         "=": 0.022571008652448654
1299                     }
1300                 },
1301                 {
1302                     "pair_idx": 397,
1303                     "dominant_op": ">",
1304                     "weights": {
1305                         "<": 0.022220740094780922,
1306                         ">": 0.023431280627846718,
1307                         "=": 0.021695461124181747
1308                     }
1309                 },
1310                 {
1311                     "pair_idx": 267,

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1312         "dominant_op": ">",
1313         "weights": {
1314             "<": 0.0227966345846653,
1315             ">": 0.023497287184000015,
1316             "=": 0.02071371302008629
1317         }
1318     },
1319 ]
1320 },
1321 {
1322     "index": 2,
1323     "steepness": 19.99909782409668,
1324     "operator_weights": {
1325         "<": 34.0,
1326         ">": 31.8,
1327         "=": 34.2
1328     },
1329     "top_comparisons": [
1330         {
1331             "pair_idx": 454,
1332             "dominant_op": "=",
1333             "weights": {
1334                 "<": 0.02500935085117817,
1335                 ">": 0.020366914570331573,
1336                 "=": 0.029470907524228096
1337             }
1338         },
1339         {
1340             "pair_idx": 135,
1341             "dominant_op": ">",
1342             "weights": {
1343                 "<": 0.024212684482336044,
1344                 ">": 0.027205176651477814,
1345                 "=": 0.019513728097081184
1346             }
1347         },
1348         {
1349             "pair_idx": 92,
1350             "dominant_op": "<",
1351             "weights": {
1352                 "<": 0.02824099175632,
1353                 ">": 0.02151164598762989,
1354                 "=": 0.02080627903342247
1355             }
1356         }
1357     ],
1358 },
1359 {
1360     "index": 3,
1361     "steepness": 19.99468421936035,
1362     "operator_weights": {
1363         "<": 32.6,
1364         ">": 33.1,
1365         "=": 34.3
1366     },
1367     "top_comparisons": [
1368         {
1369             "pair_idx": 263,
1370             "dominant_op": ">",
1371             "weights": {
1372                 "<": 0.026580549776554108,
1373                 ">": 0.030176235362887383,
1374                 "=": 0.028455618768930435
1375             }
1376         },
1377         {
1378             "pair_idx": 357,
1379             "dominant_op": "=",
1380             "weights": {

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1381         "<": 0.030333979055285454,
1382         ">": 0.018037334084510803,
1383         "=": 0.03214123845100403
1384     }
1385 },
1386 {
1387     "pair_idx": 339,
1388     "dominant_op": "<",
1389     "weights": {
1390         "<": 0.028201894834637642,
1391         ">": 0.027259066700935364,
1392         "=": 0.019259003922343254
1393     }
1394 }
1395 ]
1396 },
1397 {
1398     "index": 4,
1399     "steepness": 20.004579544067383,
1400     "operator_weights": {
1401         "<": 34.4,
1402         ">": 33.8,
1403         "=": 31.9
1404     },
1405     "top_comparisons": [
1406         {
1407             "pair_idx": 146,
1408             "dominant_op": ">",
1409             "weights": {
1410                 "<": 0.030259234830737114,
1411                 ">": 0.03076791949570179,
1412                 "=": 0.01467236690223217
1413             }
1414         },
1415         {
1416             "pair_idx": 82,
1417             "dominant_op": "<",
1418             "weights": {
1419                 "<": 0.03077125735580921,
1420                 ">": 0.01688406430184841,
1421                 "=": 0.027906792238354683
1422             }
1423         },
1424         {
1425             "pair_idx": 333,
1426             "dominant_op": "<",
1427             "weights": {
1428                 "<": 0.028150469064712524,
1429                 ">": 0.025520268827676773,
1430                 "=": 0.021647028625011444
1431             }
1432         }
1433     ]
1434 },
1435 {
1436     "index": 5,
1437     "steepness": 19.98324203491211,
1438     "operator_weights": {
1439         "<": 33.8,
1440         ">": 32.9,
1441         "=": 33.3
1442     },
1443     "top_comparisons": [
1444         {
1445             "pair_idx": 366,
1446             "dominant_op": "<",
1447             "weights": {
1448                 "<": 0.04153287038207054,
1449                 ">": 0.036108288913965225,

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1450         "=: 0.018249284476041794
1451     }
1452 },
1453 {
1454     "pair_idx": 69,
1455     "dominant_op": "=",
1456     "weights": {
1457         "<": 0.030996616929769516,
1458         ">": 0.029857879504561424,
1459         "=": 0.031154239550232887
1460     }
1461 },
1462 {
1463     "pair_idx": 391,
1464     "dominant_op": "=",
1465     "weights": {
1466         "<": 0.011602863669395447,
1467         ">": 0.028820933774113655,
1468         "=": 0.04632025212049484
1469     }
1470 }
1471 ]
1472 },
1473 {
1474     "index": 6,
1475     "steepness": 19.992597579956055,
1476     "operator_weights": {
1477         "<": 33.6,
1478         ">": 32.0,
1479         "=": 34.4
1480     },
1481     "top_comparisons": [
1482         {
1483             "pair_idx": 225,
1484             "dominant_op": "=",
1485             "weights": {
1486                 "<": 0.01787494122982025,
1487                 ">": 0.02783050760626793,
1488                 "=": 0.03050277940928936
1489             }
1490         },
1491         {
1492             "pair_idx": 119,
1493             "dominant_op": "<",
1494             "weights": {
1495                 "<": 0.02544231340289116,
1496                 ">": 0.023199481889605522,
1497                 "=": 0.024326445534825325
1498             }
1499         },
1500         {
1501             "pair_idx": 154,
1502             "dominant_op": "=",
1503             "weights": {
1504                 "<": 0.02343572862446308,
1505                 ">": 0.021771052852272987,
1506                 "=": 0.025857504457235336
1507             }
1508         }
1509     ]
1510 },
1511 {
1512     "index": 7,
1513     "steepness": 20.000003814697266,
1514     "operator_weights": {
1515         "<": 32.6,
1516         ">": 34.4,
1517         "=": 33.0
1518     },

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1519         "top_comparisons": [
1520             {
1521                 "pair_idx": 84,
1522                 "dominant_op": ">",
1523                 "weights": {
1524                     "<": 0.024619977921247482,
1525                     ">": 0.0290564876049757,
1526                     "=": 0.02532089501619339
1527                 }
1528             },
1529             {
1530                 "pair_idx": 452,
1531                 "dominant_op": "<",
1532                 "weights": {
1533                     "<": 0.028086720034480095,
1534                     ">": 0.023917103186249733,
1535                     "=": 0.025289783254265785
1536                 }
1537             },
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1539                 "pair_idx": 24,
1540                 "dominant_op": "=",
1541                 "weights": {
1542                     "<": 0.027640286833047867,
1543                     ">": 0.021651027724146843,
1544                     "=": 0.02782030776143074
1545                 }
1546             }
1547         ]
1548     },
1549     {
1550         "index": 8,
1551         "steepness": 20.007226943969727,
1552         "operator_weights": {
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1554             ">": 33.9,
1555             "=": 33.2
1556         },
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1559                 "pair_idx": 366,
1560                 "dominant_op": "<",
1561                 "weights": {
1562                     "<": 0.02792542055249214,
1563                     ">": 0.024168383330106735,
1564                     "=": 0.024179531261324883
1565                 }
1566             },
1567             {
1568                 "pair_idx": 302,
1569                 "dominant_op": ">",
1570                 "weights": {
1571                     "<": 0.02325906790792942,
1572                     ">": 0.026427095755934715,
1573                     "=": 0.023825159296393394
1574                 }
1575             },
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1577                 "pair_idx": 424,
1578                 "dominant_op": "=",
1579                 "weights": {
1580                     "<": 0.022567152976989746,
1581                     ">": 0.02492077834904194,
1582                     "=": 0.025023113936185837
1583                 }
1584             }
1585         ]
1586     },
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1588         "index": 9,
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1592             ">": 32.6,
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1601                     ">": 0.02498491480946541,
1602                     "=": 0.029924362897872925
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1607                 "dominant_op": "=",
1608                 "weights": {
1609                     "<": 0.02280467562377453,
1610                     ">": 0.02401040866971016,
1611                     "=": 0.03041614219546318
1612                 }
1613             },
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1616                 "dominant_op": "=",
1617                 "weights": {
1618                     "<": 0.027342936024069786,
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1620                     "=": 0.029299750924110413
1621                 }
1622             }
1623         ]
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1626         "index": 10,
1627         "steepness": 19.993776321411133,
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1630             ">": 32.1,
1631             "=": 34.2
1632         },
1633         "top_comparisons": [
1634             {
1635                 "pair_idx": 59,
1636                 "dominant_op": ">",
1637                 "weights": {
1638                     "<": 0.020895669236779213,
1639                     ">": 0.02928955852985382,
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1646                 "weights": {
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1651             },
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1653                 "pair_idx": 262,
1654                 "dominant_op": "<",
1655                 "weights": {
1656                     "<": 0.02819567359983921,

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1657         ">": 0.02429797127842903,
1658         "=": 0.022959519177675247
1659     }
1660 }
1661 ]
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1663 {
1664     "index": 11,
1665     "steepness": 19.985591888427734,
1666     "operator_weights": {
1667         "<": 34.4,
1668         ">": 32.5,
1669         "=": 33.0
1670     },
1671     "top_comparisons": [
1672         {
1673             "pair_idx": 365,
1674             "dominant_op": ">",
1675             "weights": {
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1677                 ">": 0.04033686965703964,
1678                 "=": 0.026891285553574562
1679             }
1680         },
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1682             "pair_idx": 30,
1683             "dominant_op": "=",
1684             "weights": {
1685                 "<": 0.028001191094517708,
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1691             "pair_idx": 369,
1692             "dominant_op": "=",
1693             "weights": {
1694                 "<": 0.009336610324680805,
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1698         }
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1700 },
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1702     "index": 12,
1703     "steepness": 20.001184463500977,
1704     "operator_weights": {
1705         "<": 34.2,
1706         ">": 33.0,
1707         "=": 32.8
1708     },
1709     "top_comparisons": [
1710         {
1711             "pair_idx": 389,
1712             "dominant_op": "=",
1713             "weights": {
1714                 "<": 0.024734992533922195,
1715                 ">": 0.027107611298561096,
1716                 "=": 0.030924709513783455
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1722             "weights": {
1723                 "<": 0.024186234921216965,
1724                 ">": 0.028401223942637444,
1725                 "=": 0.025834031403064728

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1726     }
1727     },
1728     {
1729         "pair_idx": 14,
1730         "dominant_op": "<",
1731         "weights": {
1732             "<": 0.029247231781482697,
1733             ">": 0.028390944004058838,
1734             "=": 0.01603745110332966
1735         }
1736     }
1737 ]
1738 },
1739 {
1740     "index": 13,
1741     "steepness": 20.008207321166992,
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1744         ">": 34.7,
1745         "=": 33.0
1746     },
1747     "top_comparisons": [
1748         {
1749             "pair_idx": 375,
1750             "dominant_op": ">",
1751             "weights": {
1752                 "<": 0.021979961544275284,
1753                 ">": 0.03178437054157257,
1754                 "=": 0.02359865792095661
1755             }
1756         },
1757         {
1758             "pair_idx": 160,
1759             "dominant_op": ">",
1760             "weights": {
1761                 "<": 0.021704070270061493,
1762                 ">": 0.026806388050317764,
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1767             "pair_idx": 392,
1768             "dominant_op": ">",
1769             "weights": {
1770                 "<": 0.02216557413339615,
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1772                 "=": 0.012394622899591923
1773             }
1774         }
1775     ]
1776 },
1777 {
1778     "index": 14,
1779     "steepness": 19.995954513549805,
1780     "operator_weights": {
1781         "<": 32.7,
1782         ">": 34.4,
1783         "=": 33.0
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1787             "pair_idx": 47,
1788             "dominant_op": ">",
1789             "weights": {
1790                 "<": 0.025990987196564674,
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1792                 "=": 0.02836637757718563
1793             }
1794         },

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1805             "pair_idx": 290,
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1807             "weights": {
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1812         }
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1820         ">": 32.5,
1821         "=": 34.6
1822     },
1823     "top_comparisons": [
1824         {
1825             "pair_idx": 8,
1826             "dominant_op": ">",
1827             "weights": {
1828                 "<": 0.021262403577566147,
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1835             "dominant_op": ">",
1836             "weights": {
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1840             }
1841         },
1842         {
1843             "pair_idx": 367,
1844             "dominant_op": ">",
1845             "weights": {
1846                 "<": 0.02176876738667488,
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1850         }
1851     ]
1852 }
1853 ]
1854 },
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1856     "index": 3,
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1860             "index": 0,
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1862             "operator_weights": {
1863                 "<": 32.3,

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1864         ">": 33.4,
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1866     },
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1903         "=": 33.5
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1908             "dominant_op": ">",
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1912                 "=": 0.023797890171408653
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1929                 ">": 0.023134199902415276,
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1931             }
1932         }
1933     ]

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1933     ]
1934 },
1935 {
1936     "index": 2,
1937     "steepness": 20.00238609313965,
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1941         "=": 34.6
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1945             "pair_idx": 147,
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1955             "dominant_op": "<",
1956             "weights": {
1957                 "<": 0.026899168267846107,
1958                 ">": 0.026447832584381104,
1959                 "=": 0.02158036269247532
1960             }
1961         },
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1965             "weights": {
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1970         }
1971     ]
1972 },
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1978         ">": 34.5,
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1980     },
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1985             "weights": {
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2000         {
2001             "pair_idx": 71,

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2003         "weights": {
2004             "<": 0.020652977749705315,
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2006             "=": 0.025546981021761894
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2008     }
2009 ]
2010 },
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2012     "index": 4,
2013     "steepness": 20.000131607055664,
2014     "operator_weights": {
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2016         ">": 32.2,
2017         "=": 32.8
2018     },
2019     "top_comparisons": [
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2022             "dominant_op": "=",
2023             "weights": {
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2025                 ">": 0.028038835152983665,
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2027             }
2028         },
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2030             "pair_idx": 384,
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2037         },
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2041             "weights": {
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2044                 "=": 0.02603280171751976
2045             }
2046         }
2047     ]
2048 },
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2050     "index": 5,
2051     "steepness": 19.99998664855957,
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2054         ">": 33.1,
2055         "=": 33.4
2056     },
2057     "top_comparisons": [
2058         {
2059             "pair_idx": 292,
2060             "dominant_op": "<",
2061             "weights": {
2062                 "<": 0.0284271240234375,
2063                 ">": 0.022657038643956184,
2064                 "=": 0.022481709718704224
2065             }
2066         },
2067         {
2068             "pair_idx": 47,
2069             "dominant_op": "<",
2070             "weights": {

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2071         "<": 0.027710232883691788,
2072         ">": 0.023176230490207672,
2073         "=": 0.021550588309764862
2074     }
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2077     "pair_idx": 366,
2078     "dominant_op": "<",
2079     "weights": {
2080         "<": 0.02680070511996746,
2081         ">": 0.0216884296387434,
2082         "=": 0.02283242903649807
2083     }
2084 }
2085 ]
2086 },
2087 {
2088     "index": 6,
2089     "steepness": 20.006717681884766,
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2092         ">": 32.9,
2093         "=": 32.8
2094     },
2095     "top_comparisons": [
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2097             "pair_idx": 416,
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2099             "weights": {
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2101                 ">": 0.021738462150096893,
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2122         }
2123     ]
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2126     "index": 7,
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2130         ">": 32.6,
2131         "=": 33.9
2132     },
2133     "top_comparisons": [
2134         {
2135             "pair_idx": 11,
2136             "dominant_op": "<",
2137             "weights": {
2138                 "<": 0.03974772244691849,
2139                 ">": 0.024368079379200935,

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2140         "=: 0.02194715291261673
2141     }
2142 },
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2144     "pair_idx": 91,
2145     "dominant_op": "<",
2146     "weights": {
2147         "<": 0.03174016997218132,
2148         ">": 0.017412232235074043,
2149         "=": 0.03064899705350399
2150     }
2151 },
2152 {
2153     "pair_idx": 37,
2154     "dominant_op": "=",
2155     "weights": {
2156         "<": 0.03088517114520073,
2157         ">": 0.01536605041474104,
2158         "=": 0.03234361857175827
2159     }
2160 }
2161 ]
2162 },
2163 {
2164     "index": 8,
2165     "steepness": 20.0022029876709,
2166     "operator_weights": {
2167         "<": 33.2,
2168         ">": 32.9,
2169         "=": 33.9
2170     },
2171     "top_comparisons": [
2172     {
2173         "pair_idx": 113,
2174         "dominant_op": "<",
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2180     },
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2184         "weights": {
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2186             ">": 0.01515590213239193,
2187             "=": 0.02398337796330452
2188         }
2189     },
2190     {
2191         "pair_idx": 215,
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2193         "weights": {
2194             "<": 0.024633415043354034,
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2196             "=": 0.025130929425358772
2197         }
2198     }
2199 ]
2200 },
2201 {
2202     "index": 9,
2203     "steepness": 20.003843307495117,
2204     "operator_weights": {
2205         "<": 33.6,
2206         ">": 33.3,
2207         "=": 33.2
2208     },

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2209         "top_comparisons": [
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2211                 "pair_idx": 393,
2212                 "dominant_op": "=",
2213                 "weights": {
2214                     "<": 0.02686818689107895,
2215                     ">": 0.02656356431543827,
2216                     "=": 0.028705382719635963
2217                 }
2218             },
2219             {
2220                 "pair_idx": 257,
2221                 "dominant_op": "=",
2222                 "weights": {
2223                     "<": 0.025422217324376106,
2224                     ">": 0.02510940097272396,
2225                     "=": 0.026754768565297127
2226                 }
2227             },
2228             {
2229                 "pair_idx": 250,
2230                 "dominant_op": "<",
2231                 "weights": {
2232                     "<": 0.02798621542751789,
2233                     ">": 0.02544686384499073,
2234                     "=": 0.02156468667089939
2235                 }
2236             }
2237         ]
2238     },
2239     {
2240         "index": 10,
2241         "steepness": 19.990365982055664,
2242         "operator_weights": {
2243             "<": 34.3,
2244             ">": 32.2,
2245             "=": 33.5
2246         },
2247         "top_comparisons": [
2248             {
2249                 "pair_idx": 390,
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2251                 "weights": {
2252                     "<": 0.036494724452495575,
2253                     ">": 0.04149788245558739,
2254                     "=": 0.02653481252491474
2255                 }
2256             },
2257             {
2258                 "pair_idx": 391,
2259                 "dominant_op": "=",
2260                 "weights": {
2261                     "<": 0.030588511377573013,
2262                     ">": 0.028578393161296844,
2263                     "=": 0.0349944569170475
2264                 }
2265             },
2266             {
2267                 "pair_idx": 393,
2268                 "dominant_op": "=",
2269                 "weights": {
2270                     "<": 0.013434911146759987,
2271                     ">": 0.03637988120317459,
2272                     "=": 0.037482261657714844
2273                 }
2274             }
2275         ]
2276     },
2277     {

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2278         "index": 11,
2279         "steepness": 19.996374130249023,
2280         "operator_weights": {
2281             "<": 34.0,
2282             ">": 33.0,
2283             "=": 33.1
2284         },
2285         "top_comparisons": [
2286             {
2287                 "pair_idx": 392,
2288                 "dominant_op": ">",
2289                 "weights": {
2290                     "<": 0.022357385605573654,
2291                     ">": 0.03784830868244171,
2292                     "=": 0.021851826459169388
2293                 }
2294             },
2295             {
2296                 "pair_idx": 457,
2297                 "dominant_op": ">",
2298                 "weights": {
2299                     "<": 0.024325905367732048,
2300                     ">": 0.04212259501218796,
2301                     "=": 0.00845756009221077
2302                 }
2303             },
2304             {
2305                 "pair_idx": 11,
2306                 "dominant_op": "=",
2307                 "weights": {
2308                     "<": 0.024623999372124672,
2309                     ">": 0.023554278537631035,
2310                     "=": 0.025630811229348183
2311                 }
2312             }
2313         ]
2314     },
2315     {
2316         "index": 12,
2317         "steepness": 19.990554809570312,
2318         "operator_weights": {
2319             "<": 33.3,
2320             ">": 34.3,
2321             "=": 32.5
2322         },
2323         "top_comparisons": [
2324             {
2325                 "pair_idx": 149,
2326                 "dominant_op": "<",
2327                 "weights": {
2328                     "<": 0.03636578097939491,
2329                     ">": 0.03252642601728439,
2330                     "=": 0.022441668435931206
2331                 }
2332             },
2333             {
2334                 "pair_idx": 284,
2335                 "dominant_op": ">",
2336                 "weights": {
2337                     "<": 0.02726682648062706,
2338                     ">": 0.03415995463728905,
2339                     "=": 0.02839703857898712
2340                 }
2341             },
2342             {
2343                 "pair_idx": 11,
2344                 "dominant_op": "=",
2345                 "weights": {
2346                     "<": 0.02318383753299713,

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2347         ">": 0.01535030361264944,
2348         "=": 0.04705733060836792
2349     }
2350 }
2351 ]
2352 },
2353 {
2354     "index": 13,
2355     "steepness": 19.99827766418457,
2356     "operator_weights": {
2357         "<": 32.9,
2358         ">": 34.3,
2359         "=": 32.8
2360     },
2361     "top_comparisons": [
2362         {
2363             "pair_idx": 227,
2364             "dominant_op": "=",
2365             "weights": {
2366                 "<": 0.024818625301122665,
2367                 ">": 0.024318287149071693,
2368                 "=": 0.02591441571712494
2369             }
2370         },
2371         {
2372             "pair_idx": 379,
2373             "dominant_op": "=",
2374             "weights": {
2375                 "<": 0.02490094304084778,
2376                 ">": 0.018251530826091766,
2377                 "=": 0.027951912954449654
2378             }
2379         },
2380         {
2381             "pair_idx": 245,
2382             "dominant_op": "<",
2383             "weights": {
2384                 "<": 0.02545914798974991,
2385                 ">": 0.025095975026488304,
2386                 "=": 0.01931985467672348
2387             }
2388         }
2389     ]
2390 },
2391 {
2392     "index": 14,
2393     "steepness": 20.0036678314209,
2394     "operator_weights": {
2395         "<": 32.2,
2396         ">": 34.4,
2397         "=": 33.4
2398     },
2399     "top_comparisons": [
2400         {
2401             "pair_idx": 149,
2402             "dominant_op": ">",
2403             "weights": {
2404                 "<": 0.02105291746556759,
2405                 ">": 0.027704942971467972,
2406                 "=": 0.027397023513913155
2407             }
2408         },
2409         {
2410             "pair_idx": 139,
2411             "dominant_op": ">",
2412             "weights": {
2413                 "<": 0.020645875483751297,
2414                 ">": 0.029114125296473503,
2415                 "=": 0.02352110669016838

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2416     }
2417     },
2418     {
2419         "pair_idx": 164,
2420         "dominant_op": "<",
2421         "weights": {
2422             "<": 0.02823237143456936,
2423             ">": 0.019138557836413383,
2424             "=": 0.023559099063277245
2425         }
2426     }
2427 ]
2428 },
2429 {
2430     "index": 15,
2431     "steepness": 19.99763298034668,
2432     "operator_weights": {
2433         "<": 33.2,
2434         ">": 33.3,
2435         "=": 33.4
2436     },
2437     "top_comparisons": [
2438         {
2439             "pair_idx": 89,
2440             "dominant_op": "=",
2441             "weights": {
2442                 "<": 0.023203125223517418,
2443                 ">": 0.027038611471652985,
2444                 "=": 0.029997359961271286
2445             }
2446         },
2447         {
2448             "pair_idx": 138,
2449             "dominant_op": ">",
2450             "weights": {
2451                 "<": 0.02784607745707035,
2452                 ">": 0.02820785716176033,
2453                 "=": 0.02322804555296898
2454             }
2455         },
2456         {
2457             "pair_idx": 366,
2458             "dominant_op": ">",
2459             "weights": {
2460                 "<": 0.02244698815047741,
2461                 ">": 0.030625851824879646,
2462                 "=": 0.023394949734210968
2463             }
2464         }
2465     ]
2466 }
2467 ]
2468 },
2469 {
2470     "index": 4,
2471     "n_sets": 16,
2472     "sets": [
2473         {
2474             "index": 0,
2475             "steepness": 20.003515243530273,
2476             "operator_weights": {
2477                 "<": 32.1,
2478                 ">": 36.2,
2479                 "=": 31.8
2480             },
2481             "top_comparisons": [
2482                 {
2483                     "pair_idx": 286,
2484                     "dominant_op": ">",

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2485         "weights": {
2486             "<": 0.029132749885320663,
2487             ">": 0.029642216861248016,
2488             "=": 0.027883006259799004
2489         }
2490     },
2491     {
2492         "pair_idx": 259,
2493         "dominant_op": "<",
2494         "weights": {
2495             "<": 0.02957981266081333,
2496             ">": 0.02109895832836628,
2497             "=": 0.025156918913125992
2498         }
2499     },
2500     {
2501         "pair_idx": 230,
2502         "dominant_op": ">",
2503         "weights": {
2504             "<": 0.024063652381300926,
2505             ">": 0.02685934118926525,
2506             "=": 0.024456406012177467
2507         }
2508     }
2509 ]
2510 },
2511 {
2512     "index": 1,
2513     "steepness": 20.004636764526367,
2514     "operator_weights": {
2515         "<": 32.7,
2516         ">": 34.0,
2517         "=": 33.3
2518     },
2519     "top_comparisons": [
2520         {
2521             "pair_idx": 227,
2522             "dominant_op": "<",
2523             "weights": {
2524                 "<": 0.03039596602320671,
2525                 ">": 0.02829628996551037,
2526                 "=": 0.012627361342310905
2527             }
2528         },
2529         {
2530             "pair_idx": 95,
2531             "dominant_op": "<",
2532             "weights": {
2533                 "<": 0.025556521490216255,
2534                 ">": 0.024714594706892967,
2535                 "=": 0.019848903641104698
2536             }
2537         },
2538         {
2539             "pair_idx": 271,
2540             "dominant_op": "<",
2541             "weights": {
2542                 "<": 0.026302747428417206,
2543                 ">": 0.01910869963467121,
2544                 "=": 0.0244962889701128
2545             }
2546         }
2547     ]
2548 },
2549 {
2550     "index": 2,
2551     "steepness": 20.003971099853516,
2552     "operator_weights": {
2553         "<": 32.7,

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2554         ">": 32.1,
2555         "=": 35.1
2556     },
2557     "top_comparisons": [
2558         {
2559             "pair_idx": 148,
2560             "dominant_op": ">",
2561             "weights": {
2562                 "<": 0.022852417081594467,
2563                 ">": 0.03465571999549866,
2564                 "=": 0.03091508150100708
2565             }
2566         },
2567         {
2568             "pair_idx": 257,
2569             "dominant_op": "<",
2570             "weights": {
2571                 "<": 0.034529801458120346,
2572                 ">": 0.017943523824214935,
2573                 "=": 0.02690601721405983
2574             }
2575         },
2576         {
2577             "pair_idx": 141,
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2579             "weights": {
2580                 "<": 0.024981819093227386,
2581                 ">": 0.021266641095280647,
2582                 "=": 0.025536129251122475
2583             }
2584         }
2585     ]
2586 },
2587 {
2588     "index": 3,
2589     "steepness": 20.000356674194336,
2590     "operator_weights": {
2591         "<": 33.5,
2592         ">": 33.5,
2593         "=": 33.1
2594     },
2595     "top_comparisons": [
2596         {
2597             "pair_idx": 224,
2598             "dominant_op": "=",
2599             "weights": {
2600                 "<": 0.025602614507079124,
2601                 ">": 0.029005063697695732,
2602                 "=": 0.030152274295687675
2603             }
2604         },
2605         {
2606             "pair_idx": 370,
2607             "dominant_op": "=",
2608             "weights": {
2609                 "<": 0.023879557847976685,
2610                 ">": 0.026037370786070824,
2611                 "=": 0.027047930285334587
2612             }
2613         },
2614         {
2615             "pair_idx": 432,
2616             "dominant_op": "=",
2617             "weights": {
2618                 "<": 0.027664639055728912,
2619                 ">": 0.021294310688972473,
2620                 "=": 0.027722574770450592
2621             }
2622         }

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2623     ]
2624 },
2625 {
2626     "index": 4,
2627     "steepness": 20.000743865966797,
2628     "operator_weights": {
2629         "<": 33.4,
2630         ">": 33.7,
2631         "=": 32.9
2632     },
2633     "top_comparisons": [
2634         {
2635             "pair_idx": 96,
2636             "dominant_op": "<",
2637             "weights": {
2638                 "<": 0.026713557541370392,
2639                 ">": 0.023199299350380898,
2640                 "=": 0.025091975927352905
2641             }
2642         },
2643         {
2644             "pair_idx": 400,
2645             "dominant_op": ">",
2646             "weights": {
2647                 "<": 0.02136993035674095,
2648                 ">": 0.028362227603793144,
2649                 "=": 0.022741369903087616
2650             }
2651         },
2652         {
2653             "pair_idx": 466,
2654             "dominant_op": "=",
2655             "weights": {
2656                 "<": 0.02349013276398182,
2657                 ">": 0.023988479748368263,
2658                 "=": 0.024931015446782112
2659             }
2660         }
2661     ]
2662 },
2663 {
2664     "index": 5,
2665     "steepness": 20.005779266357422,
2666     "operator_weights": {
2667         "<": 34.1,
2668         ">": 33.8,
2669         "=": 32.1
2670     },
2671     "top_comparisons": [
2672         {
2673             "pair_idx": 58,
2674             "dominant_op": ">",
2675             "weights": {
2676                 "<": 0.025302300229668617,
2677                 ">": 0.02998068369925022,
2678                 "=": 0.022283844649791718
2679             }
2680         },
2681         {
2682             "pair_idx": 299,
2683             "dominant_op": "<",
2684             "weights": {
2685                 "<": 0.028802741318941116,
2686                 ">": 0.02273433655500412,
2687                 "=": 0.025091111660003662
2688             }
2689         },
2690         {
2691             "pair_idx": 379,

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2692         "dominant_op": ">",
2693         "weights": {
2694             "<": 0.026437345892190933,
2695             ">": 0.03004412166774273,
2696             "=": 0.0198831669986248
2697         }
2698     }
2699 ]
2700 },
2701 {
2702     "index": 6,
2703     "steepness": 20.00791358947754,
2704     "operator_weights": {
2705         "<": 33.7,
2706         ">": 33.5,
2707         "=": 32.9
2708     },
2709     "top_comparisons": [
2710         {
2711             "pair_idx": 315,
2712             "dominant_op": "=",
2713             "weights": {
2714                 "<": 0.025077110156416893,
2715                 ">": 0.024854280054569244,
2716                 "=": 0.028819194063544273
2717             }
2718         },
2719         {
2720             "pair_idx": 228,
2721             "dominant_op": "<",
2722             "weights": {
2723                 "<": 0.030406462028622627,
2724                 ">": 0.02319514751434326,
2725                 "=": 0.023357421159744263
2726             }
2727         },
2728         {
2729             "pair_idx": 166,
2730             "dominant_op": "=",
2731             "weights": {
2732                 "<": 0.021676266565918922,
2733                 ">": 0.022327562794089317,
2734                 "=": 0.027776313945651054
2735             }
2736         }
2737     ]
2738 },
2739 {
2740     "index": 7,
2741     "steepness": 20.006607055664062,
2742     "operator_weights": {
2743         "<": 33.0,
2744         ">": 33.0,
2745         "=": 34.0
2746     },
2747     "top_comparisons": [
2748         {
2749             "pair_idx": 468,
2750             "dominant_op": "=",
2751             "weights": {
2752                 "<": 0.024299446493387222,
2753                 ">": 0.025295155122876167,
2754                 "=": 0.027630632743239403
2755             }
2756         },
2757         {
2758             "pair_idx": 130,
2759             "dominant_op": "=",
2760             "weights": {

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2761         "<": 0.02453918568789959,
2762         ">": 0.02109691873192787,
2763         "=": 0.02574280835688114
2764     }
2765 },
2766 {
2767     "pair_idx": 388,
2768     "dominant_op": ">",
2769     "weights": {
2770         "<": 0.019299453124403954,
2771         ">": 0.02808351442217827,
2772         "=": 0.023754196241497993
2773     }
2774 }
2775 ]
2776 },
2777 {
2778     "index": 8,
2779     "steepness": 20.00003433227539,
2780     "operator_weights": {
2781         "<": 33.3,
2782         ">": 33.8,
2783         "=": 32.9
2784     },
2785     "top_comparisons": [
2786     {
2787         "pair_idx": 127,
2788         "dominant_op": "=",
2789         "weights": {
2790             "<": 0.022911295294761658,
2791             ">": 0.024609915912151337,
2792             "=": 0.025278113782405853
2793         }
2794     },
2795     {
2796         "pair_idx": 30,
2797         "dominant_op": "<",
2798         "weights": {
2799             "<": 0.03674307465553284,
2800             ">": 0.016757555305957794,
2801             "=": 0.01690053753554821
2802         }
2803     },
2804     {
2805         "pair_idx": 316,
2806         "dominant_op": "=",
2807         "weights": {
2808             "<": 0.02017286606132984,
2809             ">": 0.02374817430973053,
2810             "=": 0.0256770309060812
2811         }
2812     }
2813 ]
2814 },
2815 {
2816     "index": 9,
2817     "steepness": 19.994354248046875,
2818     "operator_weights": {
2819         "<": 32.9,
2820         ">": 33.5,
2821         "=": 33.6
2822     },
2823     "top_comparisons": [
2824     {
2825         "pair_idx": 375,
2826         "dominant_op": ">",
2827         "weights": {
2828             "<": 0.01954091526567936,
2829             ">": 0.036612339317798615,

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2830         "=: 0.023674214258790016
2831     }
2832 },
2833 {
2834     "pair_idx": 152,
2835     "dominant_op": "<",
2836     "weights": {
2837         "<": 0.02898756042122841,
2838         ">": 0.020433202385902405,
2839         "=": 0.02576100267469883
2840     }
2841 },
2842 {
2843     "pair_idx": 131,
2844     "dominant_op": "<",
2845     "weights": {
2846         "<": 0.02760172449052334,
2847         ">": 0.024406293407082558,
2848         "=": 0.02152826637029648
2849     }
2850 }
2851 ]
2852 },
2853 {
2854     "index": 10,
2855     "steepness": 19.995235443115234,
2856     "operator_weights": {
2857         "<": 32.0,
2858         ">": 34.6,
2859         "=": 33.4
2860     },
2861     "top_comparisons": [
2862     {
2863         "pair_idx": 73,
2864         "dominant_op": "=",
2865         "weights": {
2866             "<": 0.02819826453924179,
2867             ">": 0.029992343857884407,
2868             "=": 0.03161045163869858
2869         }
2870     },
2871     {
2872         "pair_idx": 391,
2873         "dominant_op": "<",
2874         "weights": {
2875             "<": 0.04039839282631874,
2876             ">": 0.004512785002589226,
2877             "=": 0.03909015655517578
2878         }
2879     },
2880     {
2881         "pair_idx": 393,
2882         "dominant_op": "=",
2883         "weights": {
2884             "<": 0.02634277567267418,
2885             ">": 0.023277169093489647,
2886             "=": 0.032153621315956116
2887         }
2888     }
2889 ]
2890 },
2891 {
2892     "index": 11,
2893     "steepness": 20.0034236907959,
2894     "operator_weights": {
2895         "<": 34.3,
2896         ">": 31.8,
2897         "=": 33.8
2898     },

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2899         "top_comparisons": [
2900             {
2901                 "pair_idx": 232,
2902                 "dominant_op": "=",
2903                 "weights": {
2904                     "<": 0.02574237994849682,
2905                     ">": 0.021080240607261658,
2906                     "=": 0.025899924337863922
2907                 }
2908             },
2909             {
2910                 "pair_idx": 271,
2911                 "dominant_op": ">",
2912                 "weights": {
2913                     "<": 0.022950705140829086,
2914                     ">": 0.028233224526047707,
2915                     "=": 0.02057359740138054
2916                 }
2917             },
2918             {
2919                 "pair_idx": 116,
2920                 "dominant_op": "<",
2921                 "weights": {
2922                     "<": 0.025477522984147072,
2923                     ">": 0.020317019894719124,
2924                     "=": 0.02291671372950077
2925                 }
2926             }
2927         ]
2928     },
2929     {
2930         "index": 12,
2931         "steepness": 19.98894691467285,
2932         "operator_weights": {
2933             "<": 32.3,
2934             ">": 34.2,
2935             "=": 33.6
2936         },
2937         "top_comparisons": [
2938             {
2939                 "pair_idx": 392,
2940                 "dominant_op": ">",
2941                 "weights": {
2942                     "<": 0.01934739761054516,
2943                     ">": 0.041890211403369904,
2944                     "=": 0.035454362630844116
2945                 }
2946             },
2947             {
2948                 "pair_idx": 349,
2949                 "dominant_op": "<",
2950                 "weights": {
2951                     "<": 0.03602997958660126,
2952                     ">": 0.029239054769277573,
2953                     "=": 0.023427056148648262
2954                 }
2955             },
2956             {
2957                 "pair_idx": 479,
2958                 "dominant_op": ">",
2959                 "weights": {
2960                     "<": 0.034976404160261154,
2961                     ">": 0.03703015670180321,
2962                     "=": 0.015715230256319046
2963                 }
2964             }
2965         ]
2966     },
2967     {

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2968         "index": 13,
2969         "steepness": 19.996118545532227,
2970         "operator_weights": {
2971             "<": 33.0,
2972             ">": 33.3,
2973             "=": 33.7
2974         },
2975         "top_comparisons": [
2976             {
2977                 "pair_idx": 147,
2978                 "dominant_op": ">",
2979                 "weights": {
2980                     "<": 0.025406278669834137,
2981                     ">": 0.03381941094994545,
2982                     "=": 0.029999712482094765
2983                 }
2984             },
2985             {
2986                 "pair_idx": 438,
2987                 "dominant_op": ">",
2988                 "weights": {
2989                     "<": 0.028161775320768356,
2990                     ">": 0.030764997005462646,
2991                     "=": 0.027288343757390976
2992                 }
2993             },
2994             {
2995                 "pair_idx": 257,
2996                 "dominant_op": "<",
2997                 "weights": {
2998                     "<": 0.04039058834314346,
2999                     ">": 0.02545272372663021,
3000                     "=": 0.014135021716356277
3001                 }
3002             }
3003         ]
3004     },
3005     {
3006         "index": 14,
3007         "steepness": 19.990703582763672,
3008         "operator_weights": {
3009             "<": 33.0,
3010             ">": 34.4,
3011             "=": 32.6
3012         },
3013         "top_comparisons": [
3014             {
3015                 "pair_idx": 45,
3016                 "dominant_op": "=",
3017                 "weights": {
3018                     "<": 0.027523040771484375,
3019                     ">": 0.020214125514030457,
3020                     "=": 0.02902868390083313
3021                 }
3022             },
3023             {
3024                 "pair_idx": 81,
3025                 "dominant_op": "=",
3026                 "weights": {
3027                     "<": 0.02097580023109913,
3028                     ">": 0.026832208037376404,
3029                     "=": 0.027172503992915154
3030                 }
3031             },
3032             {
3033                 "pair_idx": 318,
3034                 "dominant_op": ">",
3035                 "weights": {
3036                     "<": 0.019782178103923798,

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3037         ">": 0.028249604627490044,
3038         "=": 0.02560608834028244
3039     }
3040 }
3041 ]
3042 },
3043 {
3044     "index": 15,
3045     "steepness": 20.003162384033203,
3046     "operator_weights": {
3047         "<": 33.1,
3048         ">": 34.5,
3049         "=": 32.4
3050     },
3051     "top_comparisons": [
3052         {
3053             "pair_idx": 292,
3054             "dominant_op": "<",
3055             "weights": {
3056                 "<": 0.029399774968624115,
3057                 ">": 0.02466611936688423,
3058                 "=": 0.02293987385928631
3059             }
3060         },
3061         {
3062             "pair_idx": 233,
3063             "dominant_op": ">",
3064             "weights": {
3065                 "<": 0.025411484763026237,
3066                 ">": 0.0277654267847538,
3067                 "=": 0.02208266407251358
3068             }
3069         },
3070         {
3071             "pair_idx": 78,
3072             "dominant_op": "<",
3073             "weights": {
3074                 "<": 0.028774745762348175,
3075                 ">": 0.01650867611169815,
3076                 "=": 0.025810914114117622
3077             }
3078         }
3079     ]
3080 }
3081 ]
3082 },
3083 {
3084     "index": 5,
3085     "n_sets": 16,
3086     "sets": [
3087         {
3088             "index": 0,
3089             "steepness": 20.005224227905273,
3090             "operator_weights": {
3091                 "<": 33.9,
3092                 ">": 32.0,
3093                 "=": 34.1
3094             },
3095             "top_comparisons": [
3096                 {
3097                     "pair_idx": 113,
3098                     "dominant_op": "=",
3099                     "weights": {
3100                         "<": 0.023073123767971992,
3101                         ">": 0.023731309920549393,
3102                         "=": 0.02692035213112831
3103                     }
3104                 },
3105                 {

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3106         "pair_idx": 347,
3107         "dominant_op": ">",
3108         "weights": {
3109             "<": 0.02545185573399067,
3110             ">": 0.025749264284968376,
3111             "=": 0.022128911688923836
3112         }
3113     },
3114     {
3115         "pair_idx": 394,
3116         "dominant_op": "<",
3117         "weights": {
3118             "<": 0.026431608945131302,
3119             ">": 0.024754414334893227,
3120             "=": 0.020856870338320732
3121         }
3122     }
3123 ]
3124 },
3125 {
3126     "index": 1,
3127     "steepness": 20.017051696777344,
3128     "operator_weights": {
3129         "<": 34.3,
3130         ">": 32.2,
3131         "=": 33.5
3132     },
3133     "top_comparisons": [
3134         {
3135             "pair_idx": 454,
3136             "dominant_op": "=",
3137             "weights": {
3138                 "<": 0.023809140548110008,
3139                 ">": 0.018203197047114372,
3140                 "=": 0.02846679836511612
3141             }
3142         },
3143         {
3144             "pair_idx": 422,
3145             "dominant_op": "=",
3146             "weights": {
3147                 "<": 0.02082493156194687,
3148                 ">": 0.02375161647796631,
3149                 "=": 0.02517806738615036
3150             }
3151         },
3152         {
3153             "pair_idx": 365,
3154             "dominant_op": "=",
3155             "weights": {
3156                 "<": 0.02277219481766224,
3157                 ">": 0.01940089464187622,
3158                 "=": 0.027114877477288246
3159             }
3160         }
3161     ]
3162 },
3163 {
3164     "index": 2,
3165     "steepness": 19.992206573486328,
3166     "operator_weights": {
3167         "<": 33.3,
3168         ">": 32.5,
3169         "=": 34.2
3170     },
3171     "top_comparisons": [
3172         {
3173             "pair_idx": 162,
3174             "dominant_op": "=",

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3175         "weights": {
3176             "<": 0.03329459950327873,
3177             ">": 0.028787881135940552,
3178             "=": 0.034983791410923004
3179         }
3180     },
3181     {
3182         "pair_idx": 76,
3183         "dominant_op": ">",
3184         "weights": {
3185             "<": 0.025024909526109695,
3186             ">": 0.03457939997315407,
3187             "=": 0.032640404999256134
3188         }
3189     },
3190     {
3191         "pair_idx": 450,
3192         "dominant_op": ">",
3193         "weights": {
3194             "<": 0.030129380524158478,
3195             ">": 0.03513049706816673,
3196             "=": 0.018766194581985474
3197         }
3198     }
3199 ]
3200 },
3201 {
3202     "index": 3,
3203     "steepness": 19.997770309448242,
3204     "operator_weights": {
3205         "<": 32.6,
3206         ">": 33.2,
3207         "=": 34.2
3208     },
3209     "top_comparisons": [
3210         {
3211             "pair_idx": 365,
3212             "dominant_op": "=",
3213             "weights": {
3214                 "<": 0.01793629489839077,
3215                 ">": 0.027878517284989357,
3216                 "=": 0.03442729637026787
3217             }
3218         },
3219         {
3220             "pair_idx": 2,
3221             "dominant_op": ">",
3222             "weights": {
3223                 "<": 0.026193419471383095,
3224                 ">": 0.028475526720285416,
3225                 "=": 0.02086174301803112
3226             }
3227         },
3228         {
3229             "pair_idx": 391,
3230             "dominant_op": "=",
3231             "weights": {
3232                 "<": 0.022514447569847107,
3233                 ">": 0.01902623288333416,
3234                 "=": 0.03267090395092964
3235             }
3236         }
3237     ]
3238 },
3239 {
3240     "index": 4,
3241     "steepness": 19.99616241455078,
3242     "operator_weights": {
3243         "<": 33.2,

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3244         ">": 32.9,
3245         "=": 33.9
3246     },
3247     "top_comparisons": [
3248         {
3249             "pair_idx": 68,
3250             "dominant_op": "=",
3251             "weights": {
3252                 "<": 0.029169373214244843,
3253                 ">": 0.024126825854182243,
3254                 "=": 0.031084872782230377
3255             }
3256         },
3257         {
3258             "pair_idx": 8,
3259             "dominant_op": "<",
3260             "weights": {
3261                 "<": 0.02847229316830635,
3262                 ">": 0.028249619528651237,
3263                 "=": 0.02658241055905819
3264             }
3265         },
3266         {
3267             "pair_idx": 25,
3268             "dominant_op": ">",
3269             "weights": {
3270                 "<": 0.02620520256459713,
3271                 ">": 0.02898051217198372,
3272                 "=": 0.02477569319307804
3273             }
3274         }
3275     ]
3276 },
3277 {
3278     "index": 5,
3279     "steepness": 19.989744186401367,
3280     "operator_weights": {
3281         "<": 33.4,
3282         ">": 34.0,
3283         "=": 32.6
3284     },
3285     "top_comparisons": [
3286         {
3287             "pair_idx": 363,
3288             "dominant_op": "=",
3289             "weights": {
3290                 "<": 0.02080424688756466,
3291                 ">": 0.040559351444244385,
3292                 "=": 0.04196397587656975
3293             }
3294         },
3295         {
3296             "pair_idx": 375,
3297             "dominant_op": ">",
3298             "weights": {
3299                 "<": 0.02750890888273716,
3300                 ">": 0.05414368212223053,
3301                 "=": 0.017251474782824516
3302             }
3303         },
3304         {
3305             "pair_idx": 366,
3306             "dominant_op": ">",
3307             "weights": {
3308                 "<": 0.02868787758052349,
3309                 ">": 0.052198681980371475,
3310                 "=": 0.006193717010319233
3311             }
3312         }

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3313     ]
3314 },
3315 {
3316     "index": 6,
3317     "steepness": 20.005599975585938,
3318     "operator_weights": {
3319         "<": 32.9,
3320         ">": 33.7,
3321         "=": 33.3
3322     },
3323     "top_comparisons": [
3324         {
3325             "pair_idx": 281,
3326             "dominant_op": "<",
3327             "weights": {
3328                 "<": 0.02900482341647148,
3329                 ">": 0.02768329344689846,
3330                 "=": 0.026805829256772995
3331             }
3332         },
3333         {
3334             "pair_idx": 161,
3335             "dominant_op": "=",
3336             "weights": {
3337                 "<": 0.019623734056949615,
3338                 ">": 0.026212047785520554,
3339                 "=": 0.026761561632156372
3340             }
3341         },
3342         {
3343             "pair_idx": 301,
3344             "dominant_op": "<",
3345             "weights": {
3346                 "<": 0.025388017296791077,
3347                 ">": 0.02461334876716137,
3348                 "=": 0.02182582952082157
3349             }
3350         }
3351     ]
3352 },
3353 {
3354     "index": 7,
3355     "steepness": 20.00601577758789,
3356     "operator_weights": {
3357         "<": 32.4,
3358         ">": 34.2,
3359         "=": 33.3
3360     },
3361     "top_comparisons": [
3362         {
3363             "pair_idx": 266,
3364             "dominant_op": "=",
3365             "weights": {
3366                 "<": 0.025965657085180283,
3367                 ">": 0.02798555977642536,
3368                 "=": 0.029115866869688034
3369             }
3370         },
3371         {
3372             "pair_idx": 109,
3373             "dominant_op": ">",
3374             "weights": {
3375                 "<": 0.023180412128567696,
3376                 ">": 0.028437675908207893,
3377                 "=": 0.02668398804962635
3378             }
3379         },
3380         {
3381             "pair_idx": 308,

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3382         "dominant_op": "=",
3383         "weights": {
3384             "<": 0.020920468494296074,
3385             ">": 0.026802819222211838,
3386             "=": 0.027515249326825142
3387         }
3388     }
3389 ]
3390 },
3391 {
3392     "index": 8,
3393     "steepness": 19.997602462768555,
3394     "operator_weights": {
3395         "<": 34.1,
3396         ">": 32.7,
3397         "=": 33.2
3398     },
3399     "top_comparisons": [
3400         {
3401             "pair_idx": 307,
3402             "dominant_op": "<",
3403             "weights": {
3404                 "<": 0.031770192086696625,
3405                 ">": 0.03058479353785515,
3406                 "=": 0.02813875861465931
3407             }
3408         },
3409         {
3410             "pair_idx": 27,
3411             "dominant_op": "=",
3412             "weights": {
3413                 "<": 0.02435234747827053,
3414                 ">": 0.023402482271194458,
3415                 "=": 0.029412172734737396
3416             }
3417         },
3418         {
3419             "pair_idx": 108,
3420             "dominant_op": "<",
3421             "weights": {
3422                 "<": 0.031287554651498795,
3423                 ">": 0.029226278886198997,
3424                 "=": 0.016155526041984558
3425             }
3426         }
3427     ]
3428 },
3429 {
3430     "index": 9,
3431     "steepness": 19.98790740966797,
3432     "operator_weights": {
3433         "<": 32.1,
3434         ">": 34.8,
3435         "=": 33.1
3436     },
3437     "top_comparisons": [
3438         {
3439             "pair_idx": 66,
3440             "dominant_op": "=",
3441             "weights": {
3442                 "<": 0.022175338119268417,
3443                 ">": 0.028683921322226524,
3444                 "=": 0.03060034103691578
3445             }
3446         },
3447         {
3448             "pair_idx": 265,
3449             "dominant_op": "=",
3450             "weights": {

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3451         "<": 0.029210513457655907,
3452         ">": 0.017207788303494453,
3453         "=": 0.03358006477355957
3454     }
3455 },
3456 {
3457     "pair_idx": 23,
3458     "dominant_op": "<",
3459     "weights": {
3460         "<": 0.03180839493870735,
3461         ">": 0.02262158691883087,
3462         "=": 0.025136271491646767
3463     }
3464 }
3465 ]
3466 },
3467 {
3468     "index": 10,
3469     "steepness": 19.9950008392334,
3470     "operator_weights": {
3471         "<": 32.7,
3472         ">": 34.6,
3473         "=": 32.7
3474     },
3475     "top_comparisons": [
3476     {
3477         "pair_idx": 464,
3478         "dominant_op": "=",
3479         "weights": {
3480             "<": 0.02569064311683178,
3481             ">": 0.028500650078058243,
3482             "=": 0.030670413747429848
3483         }
3484     },
3485     {
3486         "pair_idx": 395,
3487         "dominant_op": ">",
3488         "weights": {
3489             "<": 0.032793596386909485,
3490             ">": 0.034615956246852875,
3491             "=": 0.01550713274627924
3492         }
3493     },
3494     {
3495         "pair_idx": 304,
3496         "dominant_op": ">",
3497         "weights": {
3498             "<": 0.024525273591279984,
3499             ">": 0.03128852695226669,
3500             "=": 0.024025896564126015
3501         }
3502     }
3503 ]
3504 },
3505 {
3506     "index": 11,
3507     "steepness": 19.993539810180664,
3508     "operator_weights": {
3509         "<": 33.6,
3510         ">": 33.4,
3511         "=": 33.0
3512     },
3513     "top_comparisons": [
3514     {
3515         "pair_idx": 47,
3516         "dominant_op": "<",
3517         "weights": {
3518             "<": 0.035210900008678436,
3519             ">": 0.034895241260528564,

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3520         "=: 0.029137128964066505
3521     }
3522 },
3523 {
3524     "pair_idx": 25,
3525     "dominant_op": "<",
3526     "weights": {
3527         "<": 0.03663325682282448,
3528         ">": 0.028667999431490898,
3529         "=": 0.03293389454483986
3530     }
3531 },
3532 {
3533     "pair_idx": 149,
3534     "dominant_op": "<",
3535     "weights": {
3536         "<": 0.03256106749176979,
3537         ">": 0.031533099710941315,
3538         "=": 0.030627284198999405
3539     }
3540 }
3541 ]
3542 },
3543 {
3544     "index": 12,
3545     "steepness": 19.993471145629883,
3546     "operator_weights": {
3547         "<": 33.3,
3548         ">": 34.4,
3549         "=": 32.3
3550     },
3551     "top_comparisons": [
3552     {
3553         "pair_idx": 392,
3554         "dominant_op": ">",
3555         "weights": {
3556             "<": 0.030037429183721542,
3557             ">": 0.037261318415403366,
3558             "=": 0.02541530691087246
3559         }
3560     },
3561     {
3562         "pair_idx": 483,
3563         "dominant_op": ">",
3564         "weights": {
3565             "<": 0.022372189909219742,
3566             ">": 0.03145672008395195,
3567             "=": 0.024124404415488243
3568         }
3569     },
3570     {
3571         "pair_idx": 26,
3572         "dominant_op": "=",
3573         "weights": {
3574             "<": 0.023591892793774605,
3575             ">": 0.019805749878287315,
3576             "=": 0.030467556789517403
3577         }
3578     }
3579 ]
3580 },
3581 {
3582     "index": 13,
3583     "steepness": 20.00853157043457,
3584     "operator_weights": {
3585         "<": 33.0,
3586         ">": 33.9,
3587         "=": 33.1
3588     },

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3589         "top_comparisons": [
3590             {
3591                 "pair_idx": 130,
3592                 "dominant_op": "=",
3593                 "weights": {
3594                     "<": 0.025401433929800987,
3595                     ">": 0.021259071305394173,
3596                     "=": 0.028970718383789062
3597                 }
3598             },
3599             {
3600                 "pair_idx": 344,
3601                 "dominant_op": ">",
3602                 "weights": {
3603                     "<": 0.02049829252064228,
3604                     ">": 0.029082607477903366,
3605                     "=": 0.02401062846183777
3606                 }
3607             },
3608             {
3609                 "pair_idx": 451,
3610                 "dominant_op": "=",
3611                 "weights": {
3612                     "<": 0.022183701395988464,
3613                     ">": 0.022367769852280617,
3614                     "=": 0.027146294713020325
3615                 }
3616             }
3617         ]
3618     },
3619     {
3620         "index": 14,
3621         "steepness": 20.00055694580078,
3622         "operator_weights": {
3623             "<": 33.8,
3624             ">": 33.6,
3625             "=": 32.6
3626         },
3627         "top_comparisons": [
3628             {
3629                 "pair_idx": 361,
3630                 "dominant_op": "=",
3631                 "weights": {
3632                     "<": 0.025747427716851234,
3633                     ">": 0.025909116491675377,
3634                     "=": 0.032852962613105774
3635                 }
3636             },
3637             {
3638                 "pair_idx": 266,
3639                 "dominant_op": "=",
3640                 "weights": {
3641                     "<": 0.02123916707932949,
3642                     ">": 0.02980888821184635,
3643                     "=": 0.030543135479092598
3644                 }
3645             },
3646             {
3647                 "pair_idx": 263,
3648                 "dominant_op": ">",
3649                 "weights": {
3650                     "<": 0.017237862572073936,
3651                     ">": 0.03014569729566574,
3652                     "=": 0.029944662004709244
3653                 }
3654             }
3655         ]
3656     },
3657     {

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3658         "index": 15,
3659         "steepness": 20.000669479370117,
3660         "operator_weights": {
3661             "<": 33.0,
3662             ">": 33.4,
3663             "=": 33.6
3664         },
3665         "top_comparisons": [
3666             {
3667                 "pair_idx": 56,
3668                 "dominant_op": "=",
3669                 "weights": {
3670                     "<": 0.02790948562324047,
3671                     ">": 0.021460680291056633,
3672                     "=": 0.02912772074341774
3673                 }
3674             },
3675             {
3676                 "pair_idx": 289,
3677                 "dominant_op": ">",
3678                 "weights": {
3679                     "<": 0.01773759350180626,
3680                     ">": 0.029004519805312157,
3681                     "=": 0.028875146061182022
3682                 }
3683             },
3684             {
3685                 "pair_idx": 150,
3686                 "dominant_op": "=",
3687                 "weights": {
3688                     "<": 0.026119662448763847,
3689                     ">": 0.02068287879228592,
3690                     "=": 0.027521513402462006
3691                 }
3692             }
3693         ]
3694     }
3695 ]
3696 },
3697 {
3698     "index": 6,
3699     "n_sets": 16,
3700     "sets": [
3701         {
3702             "index": 0,
3703             "steepness": 19.99236297607422,
3704             "operator_weights": {
3705                 "<": 32.3,
3706                 ">": 33.2,
3707                 "=": 34.6
3708             },
3709             "top_comparisons": [
3710                 {
3711                     "pair_idx": 163,
3712                     "dominant_op": "<",
3713                     "weights": {
3714                         "<": 0.03175891563296318,
3715                         ">": 0.028528211638331413,
3716                         "=": 0.027476247400045395
3717                     }
3718                 },
3719                 {
3720                     "pair_idx": 446,
3721                     "dominant_op": ">",
3722                     "weights": {
3723                         "<": 0.024189148098230362,
3724                         ">": 0.03152765706181526,
3725                         "=": 0.03146537020802498
3726                     }
3727                 }
3728             ]
3729         }
3730     ]
3731 }

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3727     },
3728     {
3729         "pair_idx": 348,
3730         "dominant_op": "<",
3731         "weights": {
3732             "<": 0.03387954458594322,
3733             ">": 0.027894306927919388,
3734             "=": 0.01712610386312008
3735         }
3736     }
3737 ]
3738 },
3739 {
3740     "index": 1,
3741     "steepness": 20.002361297607422,
3742     "operator_weights": {
3743         "<": 33.9,
3744         ">": 33.8,
3745         "=": 32.3
3746     },
3747     "top_comparisons": [
3748         {
3749             "pair_idx": 365,
3750             "dominant_op": "<",
3751             "weights": {
3752                 "<": 0.030589634552598,
3753                 ">": 0.025819070637226105,
3754                 "=": 0.027062151581048965
3755             }
3756         },
3757         {
3758             "pair_idx": 110,
3759             "dominant_op": ">",
3760             "weights": {
3761                 "<": 0.023888811469078064,
3762                 ">": 0.026188315823674202,
3763                 "=": 0.02418646030128002
3764             }
3765         },
3766         {
3767             "pair_idx": 368,
3768             "dominant_op": "=",
3769             "weights": {
3770                 "<": 0.016653351485729218,
3771                 ">": 0.023491786792874336,
3772                 "=": 0.03293851390480995
3773             }
3774         }
3775     ]
3776 },
3777 {
3778     "index": 2,
3779     "steepness": 19.994522094726562,
3780     "operator_weights": {
3781         "<": 33.4,
3782         ">": 32.4,
3783         "=": 34.2
3784     },
3785     "top_comparisons": [
3786         {
3787             "pair_idx": 155,
3788             "dominant_op": "<",
3789             "weights": {
3790                 "<": 0.030159493908286095,
3791                 ">": 0.026482487097382545,
3792                 "=": 0.025736568495631218
3793             }
3794         },
3795         {

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3796         "pair_idx": 217,
3797         "dominant_op": "<",
3798         "weights": {
3799             "<": 0.030226049944758415,
3800             ">": 0.024640174582600594,
3801             "=": 0.023651842027902603
3802         }
3803     },
3804     {
3805         "pair_idx": 66,
3806         "dominant_op": "<",
3807         "weights": {
3808             "<": 0.029594436287879944,
3809             ">": 0.0285345409065485,
3810             "=": 0.018109168857336044
3811         }
3812     }
3813 ]
3814 },
3815 {
3816     "index": 3,
3817     "steepness": 19.98976707458496,
3818     "operator_weights": {
3819         "<": 32.9,
3820         ">": 33.8,
3821         "=": 33.3
3822     },
3823     "top_comparisons": [
3824         {
3825             "pair_idx": 286,
3826             "dominant_op": ">",
3827             "weights": {
3828                 "<": 0.02528555691242218,
3829                 ">": 0.029129525646567345,
3830                 "=": 0.028397725895047188
3831             }
3832         },
3833         {
3834             "pair_idx": 227,
3835             "dominant_op": ">",
3836             "weights": {
3837                 "<": 0.01950930617749691,
3838                 ">": 0.033632680773735046,
3839                 "=": 0.028451137244701385
3840             }
3841         },
3842         {
3843             "pair_idx": 55,
3844             "dominant_op": "=",
3845             "weights": {
3846                 "<": 0.027366124093532562,
3847                 ">": 0.024921556934714317,
3848                 "=": 0.02844071388244629
3849             }
3850         }
3851     ]
3852 },
3853 {
3854     "index": 4,
3855     "steepness": 20.0020694732666,
3856     "operator_weights": {
3857         "<": 32.7,
3858         ">": 32.0,
3859         "=": 35.2
3860     },
3861     "top_comparisons": [
3862         {
3863             "pair_idx": 84,
3864             "dominant_op": ">",

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3865         "weights": {
3866             "<": 0.0244290754199028,
3867             ">": 0.024668678641319275,
3868             "=": 0.023635489866137505
3869         }
3870     },
3871     {
3872         "pair_idx": 427,
3873         "dominant_op": "=",
3874         "weights": {
3875             "<": 0.023255346342921257,
3876             ">": 0.024463551118969917,
3877             "=": 0.024625292047858238
3878         }
3879     },
3880     {
3881         "pair_idx": 48,
3882         "dominant_op": "=",
3883         "weights": {
3884             "<": 0.024004096165299416,
3885             ">": 0.02317911759018898,
3886             "=": 0.024211136624217033
3887         }
3888     }
3889 ]
3890 },
3891 {
3892     "index": 5,
3893     "steepness": 20.005290985107422,
3894     "operator_weights": {
3895         "<": 33.7,
3896         ">": 33.9,
3897         "=": 32.3
3898     },
3899     "top_comparisons": [
3900         {
3901             "pair_idx": 129,
3902             "dominant_op": "=",
3903             "weights": {
3904                 "<": 0.02042877860367298,
3905                 ">": 0.02778150700032711,
3906                 "=": 0.029053330421447754
3907             }
3908         },
3909         {
3910             "pair_idx": 237,
3911             "dominant_op": "<",
3912             "weights": {
3913                 "<": 0.02817237190902233,
3914                 ">": 0.0230988971889019,
3915                 "=": 0.025835881009697914
3916             }
3917         },
3918         {
3919             "pair_idx": 481,
3920             "dominant_op": ">",
3921             "weights": {
3922                 "<": 0.0176225658506155,
3923                 ">": 0.029417306184768677,
3924                 "=": 0.028628017753362656
3925             }
3926         }
3927     ]
3928 },
3929 {
3930     "index": 6,
3931     "steepness": 20.0025634765625,
3932     "operator_weights": {
3933         "<": 33.1,

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3934         ">": 34.1,
3935         "=": 32.7
3936     },
3937     "top_comparisons": [
3938         {
3939             "pair_idx": 47,
3940             "dominant_op": "=",
3941             "weights": {
3942                 "<": 0.02555668354034424,
3943                 ">": 0.01902778446674347,
3944                 "=": 0.02616245485842228
3945             }
3946         },
3947         {
3948             "pair_idx": 49,
3949             "dominant_op": ">",
3950             "weights": {
3951                 "<": 0.02089553140103817,
3952                 ">": 0.024636801332235336,
3953                 "=": 0.023203708231449127
3954             }
3955         },
3956         {
3957             "pair_idx": 325,
3958             "dominant_op": ">",
3959             "weights": {
3960                 "<": 0.02005724608898163,
3961                 ">": 0.02426730841398239,
3962                 "=": 0.023723166435956955
3963             }
3964         }
3965     ]
3966 },
3967 {
3968     "index": 7,
3969     "steepness": 19.99664306640625,
3970     "operator_weights": {
3971         "<": 34.5,
3972         ">": 33.6,
3973         "=": 32.0
3974     },
3975     "top_comparisons": [
3976         {
3977             "pair_idx": 54,
3978             "dominant_op": "=",
3979             "weights": {
3980                 "<": 0.026182865723967552,
3981                 ">": 0.024111369624733925,
3982                 "=": 0.030129961669445038
3983             }
3984         },
3985         {
3986             "pair_idx": 427,
3987             "dominant_op": "<",
3988             "weights": {
3989                 "<": 0.02942327782511711,
3990                 ">": 0.025716358795762062,
3991                 "=": 0.023790962994098663
3992             }
3993         },
3994         {
3995             "pair_idx": 342,
3996             "dominant_op": "<",
3997             "weights": {
3998                 "<": 0.02806137688457966,
3999                 ">": 0.02259998582303524,
4000                 "=": 0.027264147996902466
4001             }
4002         }

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4003     ]
4004 },
4005 {
4006     "index": 8,
4007     "steepness": 20.00193977355957,
4008     "operator_weights": {
4009         "<": 32.2,
4010         ">": 33.7,
4011         "=": 34.1
4012     },
4013     "top_comparisons": [
4014         {
4015             "pair_idx": 133,
4016             "dominant_op": "<",
4017             "weights": {
4018                 "<": 0.02501479536294937,
4019                 ">": 0.024202069267630577,
4020                 "=": 0.022045498713850975
4021             }
4022         },
4023         {
4024             "pair_idx": 43,
4025             "dominant_op": "<",
4026             "weights": {
4027                 "<": 0.02693266049027443,
4028                 ">": 0.025528330355882645,
4029                 "=": 0.017567146569490433
4030             }
4031         },
4032         {
4033             "pair_idx": 335,
4034             "dominant_op": ">",
4035             "weights": {
4036                 "<": 0.020610332489013672,
4037                 ">": 0.025845622643828392,
4038                 "=": 0.02285989560186863
4039             }
4040         }
4041     ]
4042 },
4043 {
4044     "index": 9,
4045     "steepness": 20.010271072387695,
4046     "operator_weights": {
4047         "<": 34.3,
4048         ">": 32.6,
4049         "=": 33.1
4050     },
4051     "top_comparisons": [
4052         {
4053             "pair_idx": 301,
4054             "dominant_op": "=",
4055             "weights": {
4056                 "<": 0.02728816121816635,
4057                 ">": 0.02862677350640297,
4058                 "=": 0.029226448386907578
4059             }
4060         },
4061         {
4062             "pair_idx": 30,
4063             "dominant_op": "<",
4064             "weights": {
4065                 "<": 0.03368017077445984,
4066                 ">": 0.021487699821591377,
4067                 "=": 0.027425069361925125
4068             }
4069         },
4070         {
4071             "pair_idx": 440,

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4072         "dominant_op": "<",
4073         "weights": {
4074             "<": 0.02990063466131687,
4075             ">": 0.021968118846416473,
4076             "=": 0.023209061473608017
4077         }
4078     }
4079 ]
4080 },
4081 {
4082     "index": 10,
4083     "steepness": 19.993093490600586,
4084     "operator_weights": {
4085         "<": 34.0,
4086         ">": 32.6,
4087         "=": 33.3
4088     },
4089     "top_comparisons": [
4090         {
4091             "pair_idx": 461,
4092             "dominant_op": "=",
4093             "weights": {
4094                 "<": 0.027865931391716003,
4095                 ">": 0.028676437214016914,
4096                 "=": 0.029678432270884514
4097             }
4098         },
4099         {
4100             "pair_idx": 201,
4101             "dominant_op": "<",
4102             "weights": {
4103                 "<": 0.03287113085389137,
4104                 ">": 0.018134523183107376,
4105                 "=": 0.02798299863934517
4106             }
4107         },
4108         {
4109             "pair_idx": 415,
4110             "dominant_op": "=",
4111             "weights": {
4112                 "<": 0.022570613771677017,
4113                 ">": 0.024172015488147736,
4114                 "=": 0.03151262551546097
4115             }
4116         }
4117     ]
4118 },
4119 {
4120     "index": 11,
4121     "steepness": 19.997039794921875,
4122     "operator_weights": {
4123         "<": 32.7,
4124         ">": 33.9,
4125         "=": 33.4
4126     },
4127     "top_comparisons": [
4128         {
4129             "pair_idx": 395,
4130             "dominant_op": ">",
4131             "weights": {
4132                 "<": 0.012943021021783352,
4133                 ">": 0.03695661947131157,
4134                 "=": 0.03659021109342575
4135             }
4136         },
4137         {
4138             "pair_idx": 224,
4139             "dominant_op": "=",
4140             "weights": {

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4141         "<": 0.030220169574022293,
4142         ">": 0.015092429704964161,
4143         "=": 0.03177893906831741
4144     }
4145 },
4146 {
4147     "pair_idx": 75,
4148     "dominant_op": "=",
4149     "weights": {
4150         "<": 0.027044203132390976,
4151         ">": 0.01931050419807434,
4152         "=": 0.02834411710500717
4153     }
4154 }
4155 ]
4156 },
4157 {
4158     "index": 12,
4159     "steepness": 20.00179672241211,
4160     "operator_weights": {
4161         "<": 34.0,
4162         ">": 33.0,
4163         "=": 33.0
4164     },
4165     "top_comparisons": [
4166         {
4167             "pair_idx": 286,
4168             "dominant_op": "=",
4169             "weights": {
4170                 "<": 0.026651954278349876,
4171                 ">": 0.02683042176067829,
4172                 "=": 0.028055841103196144
4173             }
4174         },
4175         {
4176             "pair_idx": 89,
4177             "dominant_op": ">",
4178             "weights": {
4179                 "<": 0.023623907938599586,
4180                 ">": 0.02723868563771248,
4181                 "=": 0.02242911607027054
4182             }
4183         },
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4185             "pair_idx": 61,
4186             "dominant_op": "<",
4187             "weights": {
4188                 "<": 0.02722882479429245,
4189                 ">": 0.023697730153799057,
4190                 "=": 0.021982256323099136
4191             }
4192         }
4193     ]
4194 },
4195 {
4196     "index": 13,
4197     "steepness": 19.994258880615234,
4198     "operator_weights": {
4199         "<": 32.4,
4200         ">": 33.0,
4201         "=": 34.6
4202     },
4203     "top_comparisons": [
4204         {
4205             "pair_idx": 75,
4206             "dominant_op": "=",
4207             "weights": {
4208                 "<": 0.02843252569437027,
4209                 ">": 0.020316317677497864,

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4210         "=: 0.02958611212670803
4211     }
4212 },
4213 {
4214     "pair_idx": 271,
4215     "dominant_op": "=",
4216     "weights": {
4217         "<": 0.024970095604658127,
4218         ">": 0.02341512404382229,
4219         "=": 0.028137097135186195
4220     }
4221 },
4222 {
4223     "pair_idx": 100,
4224     "dominant_op": "=",
4225     "weights": {
4226         "<": 0.024047551676630974,
4227         ">": 0.024512041360139847,
4228         "=": 0.026463506743311882
4229     }
4230 }
4231 ]
4232 },
4233 {
4234     "index": 14,
4235     "steepness": 19.996845245361328,
4236     "operator_weights": {
4237         "<": 33.0,
4238         ">": 33.8,
4239         "=": 33.1
4240     },
4241     "top_comparisons": [
4242         {
4243             "pair_idx": 244,
4244             "dominant_op": "=",
4245             "weights": {
4246                 "<": 0.020138686522841454,
4247                 ">": 0.02666434273123741,
4248                 "=": 0.028223782777786255
4249             }
4250         },
4251         {
4252             "pair_idx": 223,
4253             "dominant_op": ">",
4254             "weights": {
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4256                 ">": 0.02937408909201622,
4257                 "=": 0.028422098606824875
4258             }
4259         },
4260         {
4261             "pair_idx": 144,
4262             "dominant_op": ">",
4263             "weights": {
4264                 "<": 0.02425839565694332,
4265                 ">": 0.026276325806975365,
4266                 "=": 0.022232526913285255
4267             }
4268         }
4269     ]
4270 },
4271 {
4272     "index": 15,
4273     "steepness": 19.99759864807129,
4274     "operator_weights": {
4275         "<": 34.1,
4276         ">": 33.0,
4277         "=": 32.9
4278     },

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4279         "top_comparisons": [
4280             {
4281                 "pair_idx": 193,
4282                 "dominant_op": ">",
4283                 "weights": {
4284                     "<": 0.02363862469792366,
4285                     ">": 0.028314316645264626,
4286                     "=": 0.026678400114178658
4287                 }
4288             },
4289             {
4290                 "pair_idx": 131,
4291                 "dominant_op": "<",
4292                 "weights": {
4293                     "<": 0.02729133516550064,
4294                     ">": 0.022995227947831154,
4295                     "=": 0.021416297182440758
4296                 }
4297             },
4298             {
4299                 "pair_idx": 265,
4300                 "dominant_op": "=",
4301                 "weights": {
4302                     "<": 0.017338264733552933,
4303                     ">": 0.02248425967991352,
4304                     "=": 0.030181167647242546
4305                 }
4306             }
4307         ]
4308     },
4309 ]
4310 },
4311 {
4312     "index": 7,
4313     "n_sets": 16,
4314     "sets": [
4315         {
4316             "index": 0,
4317             "steepness": 20.010154724121094,
4318             "operator_weights": {
4319                 "<": 34.3,
4320                 ">": 32.3,
4321                 "=": 33.4
4322             },
4323             "top_comparisons": [
4324                 {
4325                     "pair_idx": 281,
4326                     "dominant_op": ">",
4327                     "weights": {
4328                         "<": 0.02663269452750683,
4329                         ">": 0.029389962553977966,
4330                         "=": 0.02841539867222309
4331                     }
4332                 },
4333                 {
4334                     "pair_idx": 75,
4335                     "dominant_op": "<",
4336                     "weights": {
4337                         "<": 0.02933155559003353,
4338                         ">": 0.021779179573059082,
4339                         "=": 0.025046279653906822
4340                     }
4341                 },
4342                 {
4343                     "pair_idx": 273,
4344                     "dominant_op": ">",
4345                     "weights": {
4346                         "<": 0.026482289656996727,
4347                         ">": 0.02876504324376583,

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4348         "=: 0.01988827809691429
4349     }
4350 }
4351 ]
4352 },
4353 {
4354     "index": 1,
4355     "steepness": 19.998239517211914,
4356     "operator_weights": {
4357         "<": 33.4,
4358         ">": 32.3,
4359         "=": 34.3
4360     },
4361     "top_comparisons": [
4362         {
4363             "pair_idx": 389,
4364             "dominant_op": "=",
4365             "weights": {
4366                 "<": 0.019189272075891495,
4367                 ">": 0.027293171733617783,
4368                 "=": 0.02901003509759903
4369             }
4370         },
4371         {
4372             "pair_idx": 187,
4373             "dominant_op": "<",
4374             "weights": {
4375                 "<": 0.02509104646742344,
4376                 ">": 0.022206202149391174,
4377                 "=": 0.02151702530682087
4378             }
4379         },
4380         {
4381             "pair_idx": 64,
4382             "dominant_op": "<",
4383             "weights": {
4384                 "<": 0.023803770542144775,
4385                 ">": 0.021888388320803642,
4386                 "=": 0.02278442680835724
4387             }
4388         }
4389     ]
4390 },
4391 {
4392     "index": 2,
4393     "steepness": 20.000072479248047,
4394     "operator_weights": {
4395         "<": 32.5,
4396         ">": 33.1,
4397         "=": 34.4
4398     },
4399     "top_comparisons": [
4400         {
4401             "pair_idx": 449,
4402             "dominant_op": ">",
4403             "weights": {
4404                 "<": 0.0246900487691164,
4405                 ">": 0.02580796740949154,
4406                 "=": 0.024897664785385132
4407             }
4408         },
4409         {
4410             "pair_idx": 416,
4411             "dominant_op": ">",
4412             "weights": {
4413                 "<": 0.025134198367595673,
4414                 ">": 0.026942603290081024,
4415                 "=": 0.023295655846595764
4416             }
4417         }
4418     ]
4419 }

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4417     },
4418     {
4419         "pair_idx": 28,
4420         "dominant_op": "=",
4421         "weights": {
4422             "<": 0.018852872774004936,
4423             ">": 0.024241594597697258,
4424             "=": 0.02729850448668003
4425         }
4426     }
4427 ]
4428 },
4429 {
4430     "index": 3,
4431     "steepness": 20.00950813293457,
4432     "operator_weights": {
4433         "<": 32.3,
4434         ">": 34.1,
4435         "=": 33.6
4436     },
4437     "top_comparisons": [
4438         {
4439             "pair_idx": 329,
4440             "dominant_op": ">",
4441             "weights": {
4442                 "<": 0.02501615881919861,
4443                 ">": 0.027444202452898026,
4444                 "=": 0.0262162983417511
4445             }
4446         },
4447         {
4448             "pair_idx": 102,
4449             "dominant_op": ">",
4450             "weights": {
4451                 "<": 0.025462687015533447,
4452                 ">": 0.026493066921830177,
4453                 "=": 0.022349907085299492
4454             }
4455         },
4456         {
4457             "pair_idx": 229,
4458             "dominant_op": ">",
4459             "weights": {
4460                 "<": 0.02135654352605343,
4461                 ">": 0.027164924889802933,
4462                 "=": 0.02546503208577633
4463             }
4464         }
4465     ]
4466 },
4467 {
4468     "index": 4,
4469     "steepness": 19.997509002685547,
4470     "operator_weights": {
4471         "<": 32.8,
4472         ">": 34.4,
4473         "=": 32.8
4474     },
4475     "top_comparisons": [
4476         {
4477             "pair_idx": 20,
4478             "dominant_op": ">",
4479             "weights": {
4480                 "<": 0.02265894040465355,
4481                 ">": 0.02953622303903103,
4482                 "=": 0.028551647439599037
4483             }
4484         },
4485     ]

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4486         "pair_idx": 5,
4487         "dominant_op": "<",
4488         "weights": {
4489             "<": 0.0314638689160347,
4490             ">": 0.02749093621969223,
4491             "=": 0.019999682903289795
4492         }
4493     },
4494     {
4495         "pair_idx": 59,
4496         "dominant_op": ">",
4497         "weights": {
4498             "<": 0.023932991549372673,
4499             ">": 0.03144688904285431,
4500             "=": 0.02341044507920742
4501         }
4502     }
4503 ]
4504 },
4505 {
4506     "index": 5,
4507     "steepness": 20.002931594848633,
4508     "operator_weights": {
4509         "<": 34.9,
4510         ">": 32.8,
4511         "=": 32.4
4512     },
4513     "top_comparisons": [
4514         {
4515             "pair_idx": 362,
4516             "dominant_op": "=",
4517             "weights": {
4518                 "<": 0.025934768840670586,
4519                 ">": 0.022863663733005524,
4520                 "=": 0.031634218990802765
4521             }
4522         },
4523         {
4524             "pair_idx": 342,
4525             "dominant_op": "<",
4526             "weights": {
4527                 "<": 0.02937362715601921,
4528                 ">": 0.023892365396022797,
4529                 "=": 0.022891497239470482
4530             }
4531         },
4532         {
4533             "pair_idx": 465,
4534             "dominant_op": "<",
4535             "weights": {
4536                 "<": 0.02872602455317974,
4537                 ">": 0.016943935304880142,
4538                 "=": 0.028540823608636856
4539             }
4540         }
4541     ]
4542 },
4543 {
4544     "index": 6,
4545     "steepness": 19.99765396118164,
4546     "operator_weights": {
4547         "<": 32.3,
4548         ">": 33.6,
4549         "=": 34.1
4550     },
4551     "top_comparisons": [
4552         {
4553             "pair_idx": 176,
4554             "dominant_op": "<",

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4555         "weights": {
4556             "<": 0.030497830361127853,
4557             ">": 0.02306939847767353,
4558             "=": 0.028531420975923538
4559         }
4560     },
4561     {
4562         "pair_idx": 337,
4563         "dominant_op": ">",
4564         "weights": {
4565             "<": 0.02057693526148796,
4566             ">": 0.031827639788389206,
4567             "=": 0.024240665137767792
4568         }
4569     },
4570     {
4571         "pair_idx": 44,
4572         "dominant_op": "<",
4573         "weights": {
4574             "<": 0.029250307008624077,
4575             ">": 0.024884257465600967,
4576             "=": 0.022413073107600212
4577         }
4578     }
4579 ]
4580 },
4581 {
4582     "index": 7,
4583     "steepness": 20.00246238708496,
4584     "operator_weights": {
4585         "<": 33.1,
4586         ">": 32.7,
4587         "=": 34.2
4588     },
4589     "top_comparisons": [
4590         {
4591             "pair_idx": 262,
4592             "dominant_op": "=",
4593             "weights": {
4594                 "<": 0.028669942170381546,
4595                 ">": 0.018387040123343468,
4596                 "=": 0.029749510809779167
4597             }
4598         },
4599         {
4600             "pair_idx": 68,
4601             "dominant_op": ">",
4602             "weights": {
4603                 "<": 0.015971718356013298,
4604                 ">": 0.029042938724160194,
4605                 "=": 0.028230613097548485
4606             }
4607         },
4608         {
4609             "pair_idx": 141,
4610             "dominant_op": "<",
4611             "weights": {
4612                 "<": 0.026862289756536484,
4613                 ">": 0.024329083040356636,
4614                 "=": 0.02071136049926281
4615             }
4616         }
4617     ]
4618 },
4619 {
4620     "index": 8,
4621     "steepness": 20.012046813964844,
4622     "operator_weights": {
4623         "<": 32.8,

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4624         ">": 34.2,
4625         "=": 33.0
4626     },
4627     "top_comparisons": [
4628         {
4629             "pair_idx": 317,
4630             "dominant_op": "=",
4631             "weights": {
4632                 "<": 0.027148142457008362,
4633                 ">": 0.01602495275437832,
4634                 "=": 0.028024986386299133
4635             }
4636         },
4637         {
4638             "pair_idx": 337,
4639             "dominant_op": "<",
4640             "weights": {
4641                 "<": 0.025549093261361122,
4642                 ">": 0.019241318106651306,
4643                 "=": 0.024663904681801796
4644             }
4645         },
4646         {
4647             "pair_idx": 126,
4648             "dominant_op": "<",
4649             "weights": {
4650                 "<": 0.026283996179699898,
4651                 ">": 0.019849935546517372,
4652                 "=": 0.022948667407035828
4653             }
4654         }
4655     ],
4656 },
4657 {
4658     "index": 9,
4659     "steepness": 19.99797821044922,
4660     "operator_weights": {
4661         "<": 33.7,
4662         ">": 32.1,
4663         "=": 34.2
4664     },
4665     "top_comparisons": [
4666         {
4667             "pair_idx": 395,
4668             "dominant_op": ">",
4669             "weights": {
4670                 "<": 0.031779900193214417,
4671                 ">": 0.03377436101436615,
4672                 "=": 0.021442370489239693
4673             }
4674         },
4675         {
4676             "pair_idx": 5,
4677             "dominant_op": "=",
4678             "weights": {
4679                 "<": 0.02159377560019493,
4680                 ">": 0.027613388374447823,
4681                 "=": 0.027692968025803566
4682             }
4683         },
4684         {
4685             "pair_idx": 30,
4686             "dominant_op": "<",
4687             "weights": {
4688                 "<": 0.02810855023562908,
4689                 ">": 0.01985504850745201,
4690                 "=": 0.026273906230926514
4691             }
4692         }

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4693     ]
4694 },
4695 {
4696     "index": 10,
4697     "steepness": 20.004615783691406,
4698     "operator_weights": {
4699         "<": 33.2,
4700         ">": 32.8,
4701         "=": 34.0
4702     },
4703     "top_comparisons": [
4704         {
4705             "pair_idx": 297,
4706             "dominant_op": "=",
4707             "weights": {
4708                 "<": 0.02083829790353775,
4709                 ">": 0.022345226258039474,
4710                 "=": 0.028680233284831047
4711             }
4712         },
4713         {
4714             "pair_idx": 404,
4715             "dominant_op": ">",
4716             "weights": {
4717                 "<": 0.021792631596326828,
4718                 ">": 0.026536859571933746,
4719                 "=": 0.021802159026265144
4720             }
4721         },
4722         {
4723             "pair_idx": 267,
4724             "dominant_op": "=",
4725             "weights": {
4726                 "<": 0.02107338421046734,
4727                 ">": 0.024151623249053955,
4728                 "=": 0.02435535565018654
4729             }
4730         }
4731     ]
4732 },
4733 {
4734     "index": 11,
4735     "steepness": 19.995365142822266,
4736     "operator_weights": {
4737         "<": 34.5,
4738         ">": 33.5,
4739         "=": 32.0
4740     },
4741     "top_comparisons": [
4742         {
4743             "pair_idx": 10,
4744             "dominant_op": "<",
4745             "weights": {
4746                 "<": 0.028601136058568954,
4747                 ">": 0.02827099710702896,
4748                 "=": 0.021801158785820007
4749             }
4750         },
4751         {
4752             "pair_idx": 16,
4753             "dominant_op": "<",
4754             "weights": {
4755                 "<": 0.028681505471467972,
4756                 ">": 0.028101952746510506,
4757                 "=": 0.020689839497208595
4758             }
4759         },
4760         {
4761             "pair_idx": 211,

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4762         "dominant_op": "<",
4763         "weights": {
4764             "<": 0.02588784694671631,
4765             ">": 0.025452088564634323,
4766             "=": 0.02155126817524433
4767         }
4768     }
4769 ]
4770 },
4771 {
4772     "index": 12,
4773     "steepness": 20.001054763793945,
4774     "operator_weights": {
4775         "<": 32.6,
4776         ">": 33.4,
4777         "=": 34.0
4778     },
4779     "top_comparisons": [
4780         {
4781             "pair_idx": 148,
4782             "dominant_op": "=",
4783             "weights": {
4784                 "<": 0.023588893935084343,
4785                 ">": 0.022802241146564484,
4786                 "=": 0.03472187742590904
4787             }
4788         },
4789         {
4790             "pair_idx": 113,
4791             "dominant_op": ">",
4792             "weights": {
4793                 "<": 0.02642781101167202,
4794                 ">": 0.026713890954852104,
4795                 "=": 0.02335328795015812
4796             }
4797         },
4798         {
4799             "pair_idx": 146,
4800             "dominant_op": "<",
4801             "weights": {
4802                 "<": 0.027943357825279236,
4803                 ">": 0.02395511604845524,
4804                 "=": 0.02445654571056366
4805             }
4806         }
4807     ]
4808 },
4809 {
4810     "index": 13,
4811     "steepness": 20.00135040283203,
4812     "operator_weights": {
4813         "<": 33.9,
4814         ">": 32.6,
4815         "=": 33.5
4816     },
4817     "top_comparisons": [
4818         {
4819             "pair_idx": 431,
4820             "dominant_op": ">",
4821             "weights": {
4822                 "<": 0.024947255849838257,
4823                 ">": 0.02834848128259182,
4824                 "=": 0.02712007611989975
4825             }
4826         },
4827         {
4828             "pair_idx": 156,
4829             "dominant_op": "=",
4830             "weights": {

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4831         "<": 0.02463732473552227,
4832         ">": 0.021979520097374916,
4833         "=": 0.025538919493556023
4834     }
4835 },
4836 {
4837     "pair_idx": 211,
4838     "dominant_op": "=",
4839     "weights": {
4840         "<": 0.02166169136762619,
4841         ">": 0.023421963676810265,
4842         "=": 0.02581137605011463
4843     }
4844 }
4845 ]
4846 },
4847 {
4848     "index": 14,
4849     "steepness": 20.002639770507812,
4850     "operator_weights": {
4851         "<": 34.5,
4852         ">": 31.8,
4853         "=": 33.6
4854     },
4855     "top_comparisons": [
4856         {
4857             "pair_idx": 131,
4858             "dominant_op": "<",
4859             "weights": {
4860                 "<": 0.027405856177210808,
4861                 ">": 0.025174126029014587,
4862                 "=": 0.023458382114768028
4863             }
4864         },
4865         {
4866             "pair_idx": 93,
4867             "dominant_op": "<",
4868             "weights": {
4869                 "<": 0.027405882254242897,
4870                 ">": 0.02575474977493286,
4871                 "=": 0.019857047125697136
4872             }
4873         },
4874         {
4875             "pair_idx": 202,
4876             "dominant_op": "<",
4877             "weights": {
4878                 "<": 0.025495385751128197,
4879                 ">": 0.0231732539832592,
4880                 "=": 0.023554127663373947
4881             }
4882         }
4883     ]
4884 },
4885 {
4886     "index": 15,
4887     "steepness": 20.00558090209961,
4888     "operator_weights": {
4889         "<": 34.6,
4890         ">": 31.7,
4891         "=": 33.7
4892     },
4893     "top_comparisons": [
4894         {
4895             "pair_idx": 101,
4896             "dominant_op": "=",
4897             "weights": {
4898                 "<": 0.020593035966157913,
4899                 ">": 0.02590487152338028,

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4900         "=: 0.02737530693411827
4901     }
4902 },
4903 {
4904     "pair_idx": 188,
4905     "dominant_op": "<",
4906     "weights": {
4907         "<": 0.025039568543434143,
4908         ">": 0.02372686006128788,
4909         "=": 0.023751752451062202
4910     }
4911 },
4912 {
4913     "pair_idx": 364,
4914     "dominant_op": "<",
4915     "weights": {
4916         "<": 0.02867908962070942,
4917         ">": 0.02267426624894142,
4918         "=": 0.019655680283904076
4919     }
4920 }
4921 ]
4922 }
4923 ]
4924 }
4925 ]
4926 },
4927 "output_head": {
4928     "sensitivities": {
4929         "target_call_offset": {
4930             "market_physics_h": 0.41818538308143616,
4931             "portfolio_v": 0.21843554079532623,
4932             "momentum_m": 0.31555840373039246,
4933             "regime_r": 0.20850121974945068
4934         },
4935         "target_put_offset": {
4936             "market_physics_h": 0.4360724985599518,
4937             "portfolio_v": 0.24388915300369263,
4938             "momentum_m": 0.2763805091381073,
4939             "regime_r": 0.2660418748855591
4940         },
4941         "target_wing_width": {
4942             "market_physics_h": 0.5541657209396362,
4943             "portfolio_v": 0.317993700504303,
4944             "momentum_m": 0.3907780349254608,
4945             "regime_r": 0.32660138607025146
4946         },
4947         "target_dte": {
4948             "market_physics_h": 0.4183441400527954,
4949             "portfolio_v": 0.2021690309047699,
4950             "momentum_m": 0.28711408376693726,
4951             "regime_r": 0.1755628138780594
4952         },
4953         "was_profitable": {
4954             "market_physics_h": 0.42624855041503906,
4955             "portfolio_v": 0.19359971582889557,
4956             "momentum_m": 0.2834716737270355,
4957             "regime_r": 0.15474046766757965
4958         },
4959         "realized_roi": {
4960             "market_physics_h": 0.36879757046699524,
4961             "portfolio_v": 0.2159566879272461,
4962             "momentum_m": 0.25968024134635925,
4963             "regime_r": 0.20383617281913757
4964         },
4965         "realized_max_loss": {
4966             "market_physics_h": 0.3892645537853241,
4967             "portfolio_v": 0.20923279225826263,
4968             "momentum_m": 0.24894870817661285,

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4969         "regime_r": 0.2032037228345871
4970     },
4971     "confidence_target": {
4972         "market_physics_h": 0.5136378407478333,
4973         "portfolio_v": 0.22590547800064087,
4974         "momentum_m": 0.3967027962207794,
4975         "regime_r": 0.24534553289413452
4976     },
4977     "entry_target": {
4978         "market_physics_h": 0.42828240990638733,
4979         "portfolio_v": 0.20505115389823914,
4980         "momentum_m": 0.30343127250671387,
4981         "regime_r": 0.18107132613658905
4982     },
4983     "exit_target": {
4984         "market_physics_h": 0.3872460126876831,
4985         "portfolio_v": 0.22109581530094147,
4986         "momentum_m": 0.2628363072872162,
4987         "regime_r": 0.2143009901046753
4988     }
4989 }
4990 },
4991 "rules": [
4992     "RULE 1 (Vol): Enter if IVR > 75.00",
4993     "RULE 2 (Liq): Block if Spread/Price > 1.0278%",
4994     "RULE 3 (Trend): Signal Reversal if RSI < 25.00 and delta_RSI < 0",
4995     "RULE 4 (Gap): Guard if 1m price jump > 1.05%",
4996     "RULE 5 (Greeks): Hedge if |Gamma| > 0.0110",
4997     "SUPER_SET 0: 16 logic sets",
4998     " \u2514\u2500 SET 0: operators: < 32.7% | > 35.0% | = 32.4%",
4999     " \u2514\u2500 Pair 16: < dominates (<0.03, >0.02, =0.03)",
5000     " \u2514\u2500 Pair 430: < dominates (<0.03, >0.03, =0.02)",
5001     " \u2514\u2500 SET 1: operators: < 33.1% | > 32.6% | = 34.3%",
5002     " \u2514\u2500 Pair 349: < dominates (<0.03, >0.03, =0.02)",
5003     " \u2514\u2500 Pair 357: < dominates (<0.03, >0.01, =0.02)",
5004     "SUPER_SET 1: 16 logic sets",
5005     " \u2514\u2500 SET 0: operators: < 33.5% | > 33.8% | = 32.8%",
5006     " \u2514\u2500 Pair 176: < dominates (<0.03, >0.03, =0.03)",
5007     " \u2514\u2500 Pair 431: > dominates (<0.02, >0.03, =0.03)",
5008     " \u2514\u2500 SET 1: operators: < 35.1% | > 32.9% | = 32.0%",
5009     " \u2514\u2500 Pair 374: > dominates (<0.03, >0.03, =0.02)",
5010     " \u2514\u2500 Pair 438: < dominates (<0.03, >0.02, =0.03)"
5011 ]
5012 }

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